Lesson 1 — Extra Practice

What Is a System of Linear Equations? Solving by Graphing

Part A — Does the Point Satisfy *Both* Equations?

1. Determine whether the given ordered pair is the solution to the system. Substitute into **both** equations and show the arithmetic — some of these points sit on one line but not the other.

(a) $(2, -1)$ for $y = 3x - 7$ and $y = -2x + 3$.

(b) $(3, 2)$ for $y = x - 1$ and $y = -2x + 9$.

(c) $(-2, 5)$ for $2x + y = 1$ and $y = -x + 4$.

2. Determine whether the given ordered pair is the solution to the system. Both equations are in standard form, so substitute directly — there is no need to rearrange.

(a) $(4, -3)$ for $3x + 2y = 6$ and $x - y = 7$.

(b) $(-1, -4)$ for $2x - y = 2$ and $5x + 3y = -17$.

(c) $(5, 1)$ for $x + 4y = 9$ and $3x - 2y = 12$.

3. Exactly one of the three points below is the solution to the system $2x + y = 7$ and $y = x - 2$. Test each point in both equations. State which point is the solution, and for each of the other two say *which* equation it satisfies and which one it fails.

(a) $(1, -1)$

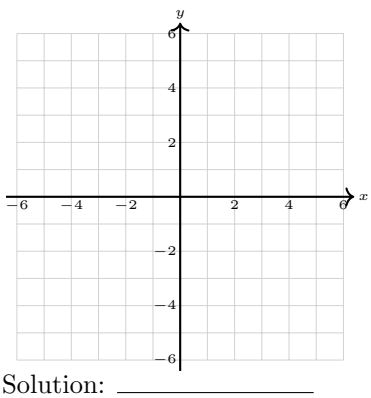
(b) $(3, 1)$

(c) $(4, -1)$

Part B — Solving by Graphing

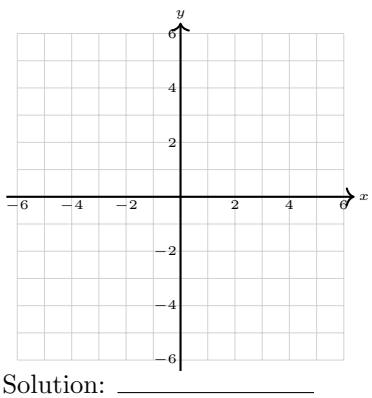
4. Graph both lines of each system by hand on the grid provided, then state the solution as an ordered pair. Each system has a **fractional slope**, so count the rise and the run one square at a time.

(a) $y = \frac{1}{2}x + 1$ and $y = -x + 4$



Solution: _____

(b) $y = \frac{2}{3}x - 2$ and $y = -\frac{1}{3}x + 1$

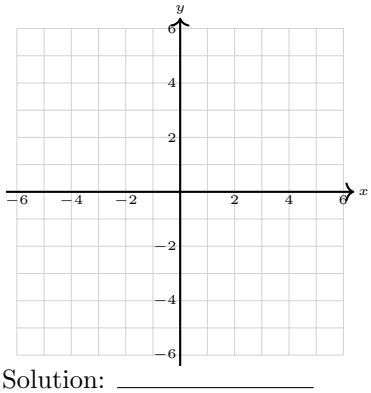


Solution: _____

5. In each system **both** equations must be rewritten in slope-intercept form before you can graph them. Rewrite each one, graph both lines, and state the solution.

(a) $2x + y = 4$ and $3x - y = 6$

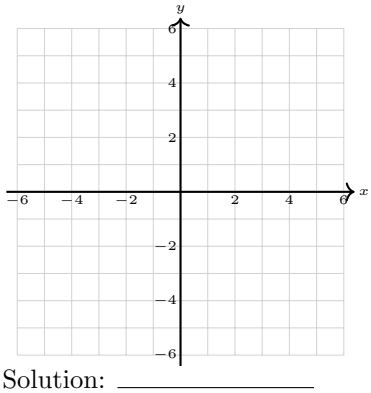
Rewritten: _____ and _____



Solution: _____

(b) $3x + 2y = 6$ and $x - y = -3$

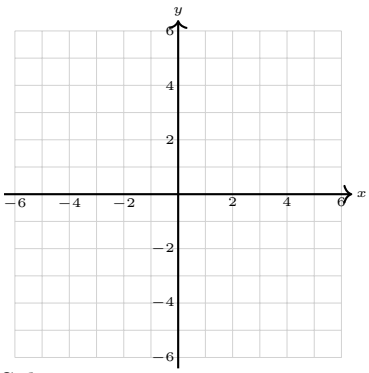
Rewritten: _____ and _____



Solution: _____

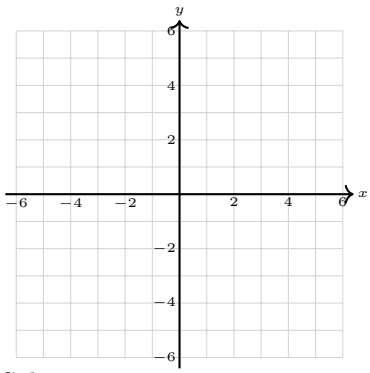
6. A line of the form $x = a$ is vertical and a line of the form $y = b$ is horizontal. Graph each system and state the solution — no slope counting is needed, but read the two equations carefully before you draw.

(a) $x = -5$ and $y = 3$



Solution: _____

(b) $3x = 9$ and $2y = -8$

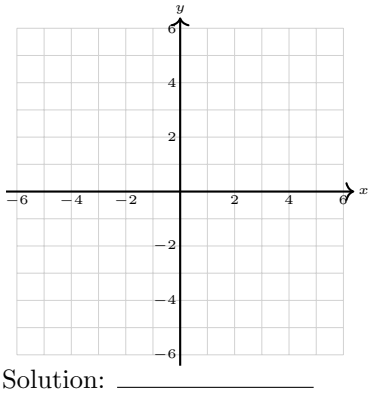


Solution: _____

7–8. Rewrite each equation in slope-intercept form, graph both lines, state the solution, and then **check that solution in both original equations**.

7. $x + 3y = 9$ and $2x - y = 4$

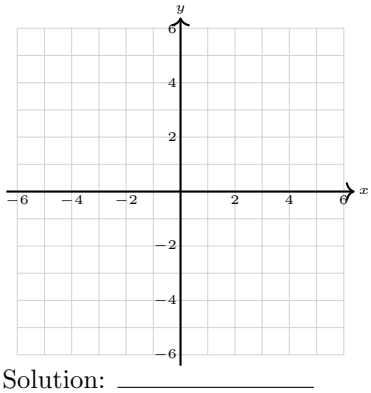
Rewritten: _____ and _____



Solution: _____

8. $y = -3x + 5$ and $x - y = 3$

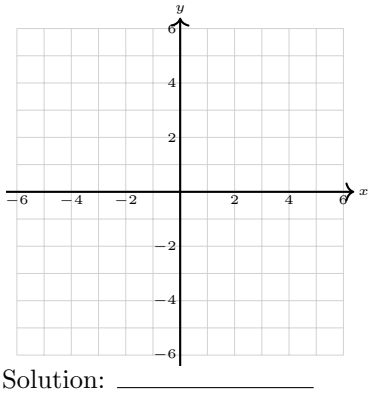
Rewritten: _____



Solution: _____

Check for #7 in $x + 3y = 9$ and $2x - y = 4$:

Check for #8 in $y = -3x + 5$ and $x - y = 3$:



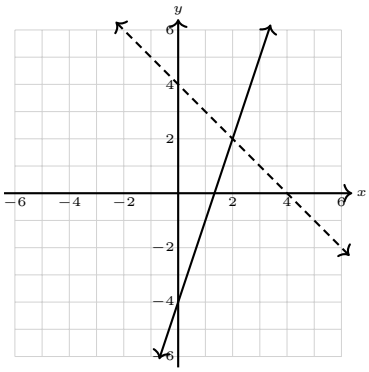
Solution: _____

9. Consider the system $y = \frac{1}{4}x + 3$ and $y = -x - 2$.
(a) The first line rises only 1 square for every 4 squares to the right. Starting at its y-intercept, list two more points on that line.

(b) Graph both lines on the grid at left and state the solution.
(c) Verify your solution in both equations.

Part C — Reading a Graph, and Finding the Error

10. The system $y = 3x - 4$ (solid) and $y = -x + 4$ (dashed) is graphed below.



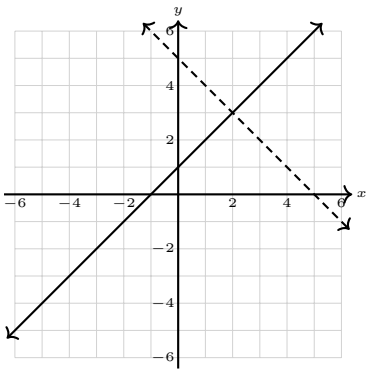
(a) State the point of intersection as an ordered pair.

(b) Substitute that pair into $y = 3x - 4$ and show whether it is true.

(c) Substitute the same pair into $y = -x + 4$ and show whether it is true.

(d) Explain what parts (b) and (c) together confirm.

11. **Error analysis.** Devon graphed the system $y = x + 1$ (solid) and $y = -x + 5$ (dashed) below and wrote the solution as $(3, 4)$.



(a) Substitute $(3, 4)$ into $y = x + 1$. Is it true?

(b) Substitute $(3, 4)$ into $y = -x + 5$. Is it true?

(c) Devon's point lies on one of the two lines but not on the other. Which line is it on, and how many squares off is his reading?

(d) State the correct solution.

12. **Error analysis.** Priya says $(0, 3)$ is the solution to the system $y = 2x + 3$ and $y = -x + 9$. Her work reads: "$y = 2x + 3$, and $2(0) + 3 = 3$ ✓, so $(0, 3)$ is the solution."

(a) What step did Priya leave out? Carry it out and show the result.

(b) Find the correct solution, and state in one sentence what a point that satisfies only one equation tells you about the graph.

Part D — Working Backwards, a Context, and a Stretch

13. In each system below the solution is given but one coefficient is missing. Use the solution — it must satisfy **both** equations — to find the missing value. Start with the equation that has no unknown coefficient and confirm the point really does satisfy it.

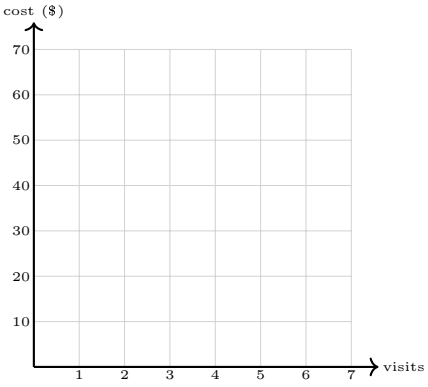
(a) Find the value of m for which $y = 2x - 5$ and $y = mx + 7$ have solution $(3, 1)$.

(b) Find the value of k for which $y = -x + 6$ and $y = kx - 4$ have solution $(5, 1)$.

(c) Find the value of b for which $3x + by = 12$ and $y = x - 1$ have solution $(2, 1)$.

(d) Find the value of a for which $x + y = 7$ and $ax - y = 5$ have solution $(4, 3)$.

14. A climbing gym sells two memberships. Under **Plan A** each visit costs \$10 and there is no monthly fee, so the monthly cost is $y = 10x$, where x is the number of visits. Under **Plan B** there is a \$20 monthly fee plus \$5 per visit, so $y = 20 + 5x$.



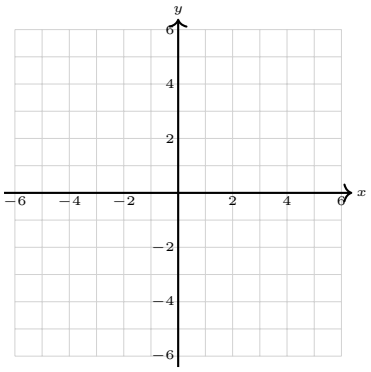
(a) On the grid at left, one square across is one visit and one square up is \$10. Graph both plans.
(b) State the point of intersection and check it in both equations.

(c) Explain in a sentence what that point means for a member of this gym.

(d) Which plan is cheaper for someone who visits 6 times a month? Use the graph to justify your answer.

15. **Three lines.** Three lines pass through a single common point only if the solution of *any two* of them also satisfies the third. For each trio, find where the first two lines meet, then test that point in the third equation.
(a) $y = 2x - 1$, $y = -x + 5$, $y = x + 1$ (b) $y = -2x + 6$, $y = x - 3$, $y = 3x - 10$

16. Write a system of two linear equations, each in the form $y = mx + b$ and with *different* slopes, whose solution is $(-2, 3)$. Then verify your system by substituting $(-2, 3)$ into both of your equations.



17. **Stretch.** Graph the three lines $y = x + 2$, $y = -x + 4$ and $y = -2$ on the grid at left. Taken two at a time they form three different systems, and the three solutions are the corners of a triangle.
(a) State the three points of intersection.

(b) The horizontal side of the triangle lies along $y = -2$. State its length in units, and the height of the triangle above it.

(c) Determine the area of the triangle.

Answer Key. 1. (a) Yes — $3(2) - 7 = -1$ ✓ and $-2(2) + 3 = -1$ ✓. (b) No — $3 - 1 = 2$ ✓ but $-2(3) + 9 = 3 \neq 2$; it lies on the first line only. (c) No — $2(-2) + 5 = 1$ ✓ but $-(-2) + 4 = 6 \neq 5$; it lies on the first line only. **2.** (a) Yes — $3(4) + 2(-3) = 6$ ✓ and $4 - (-3) = 7$ ✓. (b) Yes — $2(-1) - (-4) = 2$ ✓ and $5(-1) + 3(-4) = -17$ ✓. (c) No — $5 + 4(1) = 9$ ✓ but $3(5) - 2(1) = 13 \neq 12$. **3.** (b) $(3, 1)$ is the solution: $2(3) + 1 = 7$ ✓ and $3 - 2 = 1$ ✓. (a) $(1, -1)$ satisfies $y = x - 2$ only ($1 - 2 = -1$ ✓) and fails $2x + y = 7$ ($2 + (-1) = 1 \neq 7$). (c) $(4, -1)$ satisfies $2x + y = 7$ only ($8 - 1 = 7$ ✓) and fails $y = x - 2$ ($4 - 2 = 2 \neq -1$). **4.** (a) $(2, 2)$ (b) $(3, 0)$. **5.** (a) $y = -2x + 4$ and $y = 3x - 6$; $(2, 0)$. (b) $y = -\frac{3}{2}x + 3$ and $y = x + 3$; $(0, 3)$ — the lines meet on the y-axis. **6.** (a) $(-5, 3)$ (b) $3x = 9$ is the vertical line $x = 3$ and $2y = -8$ is the horizontal line $y = -4$, so $(3, -4)$. **7.** $y = -\frac{1}{3}x + 3$ and $y = 2x - 4$; $(3, 2)$. Check: $3 + 3(2) = 9$ ✓, $2(3) - 2 = 4$ ✓. **8.** $x - y = 3$ becomes $y = x - 3$; $(2, -1)$. Check: $-3(2) + 5 = -1$ ✓, $2 - (-1) = 3$ ✓. **9.** (a) From $(0, 3)$: $(4, 4)$ and $(-4, 2)$ (also $(8, 5)$, $(-8, 1)$). (b) $(-4, 2)$. (c) $\frac{1}{4}(-4) + 3 = -1 + 3 = 2$ ✓ and $-(-4) - 2 = 2$ ✓. **10.** (a) $(2, 2)$ (b) $3(2) - 4 = 2$ ✓ (c) $-(2) + 4 = 2$ ✓ (d) The point lies on *both* lines, so it satisfies both equations and is therefore the solution to the system. **11.** (a) Yes — $3 + 1 = 4$ ✓. (b) No — $-3 + 5 = 2 \neq 4$. (c) It lies on $y = x + 1$ (the solid line); he slid one square up and one square right along that line, past the crossing point. (d) $(2, 3)$. **12.** (a) She never substituted into the second equation: $-0 + 9 = 9 \neq 3$, so $(0, 3)$ fails $y = -x + 9$ and is not the solution. (b) The solution is $(2, 7)$ (a table of values: at $x = 2$ both give 7); a point satisfying only one equation lies on that one line but not on the other, so it is not a crossing point. **13.** (a) $(3, 1)$ satisfies $y = 2x - 5$ since $2(3) - 5 = 1$; then $1 = 3m + 7$, so $m = -2$. (b) $-5 + 6 = 1$ ✓; then $1 = 5k - 4$, so $k = 1$. (c) $y = x - 1$ gives $2 - 1 = 1$ ✓; then $3(2) + b(1) = 12$, so $b = 6$. (d) $4 + 3 = 7$ ✓; then $4a - 3 = 5$, so $a = 2$. **14.** (a) Plan A rises one square up for each square right from the origin; Plan B starts 2 squares up ($20) and rises one square for every two to the right. (b) $(4, 40)$: $10(4) = 40$ ✓ and $20 + 5(4) = 40$ ✓. (c) At 4 visits a month the two plans cost the same, $40. (d) At 6 visits Plan B is lower on the graph — $50 versus $60 — so Plan B is cheaper. (Plan A is cheaper for fewer than 4 visits.) **15.** (a) Yes. The first two lines cross at $(2, 3)$ (both give $y = 3$ at $x = 2$); testing $(2, 3)$ in $y = x + 1$ gives $2 + 1 = 3$ ✓, so all three pass through $(2, 3)$. (b) No. The first two lines cross at $(3, 0)$ (both give $y = 0$ at $x = 3$); testing $(3, 0)$ in $y = 3x - 10$ gives $3(3) - 10 = -1 \neq 0$, so the third line misses that point and the three lines do not share a common point. **16.** Many answers; any two lines with different slopes through $(-2, 3)$ work, for example $y = x + 5$ and $y = -x + 1$: $-2 + 5 = 3$ ✓ and $-(-2) + 1 = 3$ ✓. **17.** (a) $y = x + 2$ and $y = -x + 4$ meet at $(1, 3)$; $y = x + 2$ and $y = -2$ meet at $(-4, -2)$; $y = -x + 4$ and $y = -2$ meet at $(6, -2)$. (b) Base from $x = -4$ to $x = 6$ is 10 units; the height from $y = -2$ up to $y = 3$ is 5 units. (c) $A = \frac{1}{2}(10)(5) = 25$ square units.

Lesson 2 — Extra Practice

Classifying Systems: One Solution, No Solution, or Infinite Solutions

Part A — Compare m and b, Don't Graph

1. Each system is already in slope y-intercept form. State whether it has **one solution**, **no solution**, or **infinitely many solutions**. Watch for equations that are written differently but say the same thing.

(a) $y = \frac{2}{3}x - 4$
$y = \frac{2}{3}x + 1$

(b) $y = -5x + 2$
$y = -\frac{1}{5}x + 2$

(c) $y = \frac{3}{4}x - 2$
$y = \frac{6}{8}x - 2$

(d) $y = -\frac{x}{2} + 3$
$y = -0.5x + 3$

(e) $y = \frac{5}{2}x + 1$
$y = 2.5x - 1$

(f) $y = 7$
$x = 7$

2. The slopes below only match *after* you simplify them. Simplify each slope and each y-intercept, then classify the system.

(a) $y = \frac{4}{6}x + 3$
$y = \frac{2}{3}x + 3$

(b) $y = \frac{9}{12}x - 5$
$y = \frac{3}{4}x + 5$

(c) $y = \frac{10}{4}x + \frac{1}{2}$
$y = 2.5x + 0.5$

Part B — General Form: Rearrange First

3. Each equation is in standard form, $Ax + By = C$. Rewrite **both** equations as $y = mx + b$, then classify the system. In parts (a) and (b) the two equations are very nearly the same — look carefully at the constant.

(a) $2x + 4y = 10$ and $x + 2y = 5$

(b) $2x + 4y = 10$ and $x + 2y = 7$

(c) $4x - y = 7$ and $x + 2y = 4$

(d) $6x - 9y = 12$ and $2x - 3y = 5$

4. Clear the fractions or decimals first, then rewrite each equation as $y = mx + b$ and classify the system.

(a) $\frac{1}{2}x + \frac{1}{3}y = 1$ and $3x + 2y = 6$

(b) $\frac{2}{3}x - y = 4$ and $4x - 6y = 18$

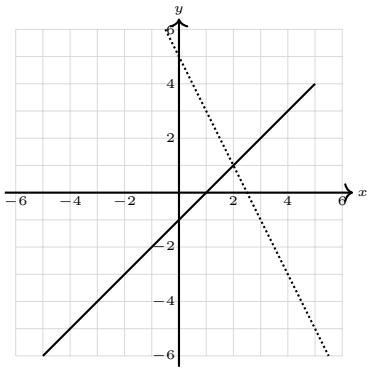
(c) $\frac{3}{4}x + y = 2$ and $3x + 2y = 10$

(d) Some of the systems in questions 3 and 4 are a single equation multiplied by a constant. Identify every such pair and say what multiplying by a constant does to m and to b.

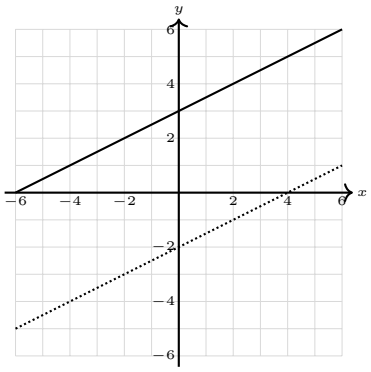
Part C — Reading a Graph

5. Classify each system below from its graph. In these diagrams two distinct lines are drawn as one **solid** line and one **dotted** line; a **coincident** pair is drawn as a thick grey line with a dashed line lying exactly on top of it, so you see only one line with a dashed stripe running along it. For each graph, state the number of solutions, and also give the solution as an ordered pair if there is exactly one.

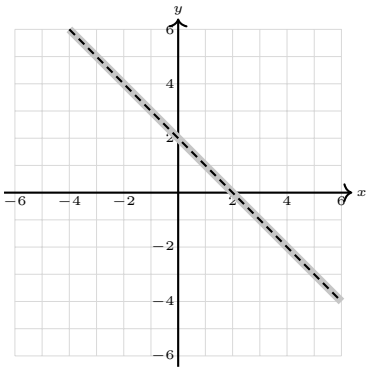
(a)



(b)



(c)



Part D — Find the Value of k

6–10. For each system, determine the value of k (if one exists) that makes the system have (a) **no solution**, (b) **infinitely many solutions**, and (c) **exactly one solution**. One of the three cases is impossible in every question below — say which one, and explain why it cannot happen.

6. $y = kx - 4$ and $y = 2x + 1$
(a) no solution: $k =$ _____
(b) infinitely many: $k =$ _____
(c) one solution: $k =$ _____

7. $y = 5x + k$ and $y = 5x - 3$
(a) no solution: $k =$ _____
(b) infinitely many: $k =$ _____
(c) one solution: $k =$ _____

8. $kx + 3y = 9$ and $2x + y = 3$
(a) no solution: $k =$ _____
(b) infinitely many: $k =$ _____
(c) one solution: $k =$ _____

9. $8x + 4y = 20$ and $2x + y = k$
(a) no solution: $k =$ _____
(b) infinitely many: $k =$ _____
(c) one solution: $k =$ _____

10. $12x - ky = 24$ and $4x - 3y = 10$. Find k for each case, then **substitute your value back in** and rewrite both equations as $y = mx + b$ to check that the case really occurs.

Part E — Error Analysis and Reasoning

11. Marcus is given the system $6x - 4y = 10$ and $3x - 2y = 5$. He writes: "The two equations have different numbers in them, so they must be different lines. Different lines cross once, so the system has one solution." Identify the flaw in his reasoning, show the correct work, and state the correct classification.

12. Can a system of two linear equations have **exactly two** solutions? Explain your answer using what you know about how many lines pass through two given points.

13. Suppose you write two linear equations as $y = mx + b$ and find that the slopes are **different**. Is it possible for the system to have **zero** solutions? Explain why or why not, and say whether you need the y-intercepts at all to answer.

14. The equation $3x - 6y = 12$ is given. Write a *second* equation, in standard form $Ax + By = C$, that pairs with it to make a system with (a) no solution, (b) exactly one solution, (c) infinitely many solutions.

Part F — Two Situations in Words

15. Gym A charges a \$40 joining fee plus \$25 per month. Gym B charges a \$75 joining fee plus \$25 per month. Write an equation for each gym's total cost y after x months, classify the system, and explain in one sentence what your classification means about the two gyms' costs.

16. A canoe rental is described two ways. **Clerk 1:** "It's \$30 to start plus \$12 an hour." **Clerk 2:** "Three times your total is \$36 per hour plus \$90." Write an equation for each description, classify the system, and explain what the classification tells you about the two clerks.

Part G — Stretch

17. Three lines are given:

$$L_1:\ y = 2x - 1 \qquad L_2:\ x + y = 5 \qquad L_3:\ 4x - 2y = 14$$

(a) Write all three in the form $y = mx + b$. Then classify each of the three pairs L_1 & L_2, L_1 & L_3, L_2 & L_3.

(b) Find the solution of every pair that has one. Is there a single point that lies on **all three** lines? Explain how your answers to (a) make this impossible without any further algebra.

Answer Key. 1. (a) no solution ($m = \frac{2}{3}$ both, b differs) (b) one solution ($-5 \neq -\frac{1}{5}$) (c) infinitely many ($\frac{6}{8} = \frac{3}{4}$, same b) (d) infinitely many ($-\frac{1}{2} = -0.5$, same b) (e) no solution ($\frac{5}{2} = 2.5$, b differs) (f) one solution, $(7, 7)$ — a horizontal line and a vertical line always meet exactly once. **2.** (a) $m = \frac{2}{3}$ both, $b = 3$ both: infinitely many (b) $m = \frac{3}{4}$ both, $b = -5$ vs 5: no solution (c) $m = \frac{5}{2}$ both, $b = \frac{1}{2}$ both: infinitely many. **3.** (a) $y = -\frac{1}{2}x + \frac{5}{2}$ and $y = -\frac{1}{2}x + \frac{5}{2}$: infinitely many (b) $y = -\frac{1}{2}x + \frac{5}{2}$ and $y = -\frac{1}{2}x + \frac{7}{2}$: no solution (c) $y = 4x - 7$ and $y = -\frac{1}{2}x + 2$: one solution (d) $y = \frac{2}{3}x - \frac{4}{3}$ and $y = \frac{2}{3}x - \frac{5}{3}$: no solution. **4.** (a) $\times 6$ gives $3x + 2y = 6$, identical: $y = -\frac{3}{2}x + 3$ twice, infinitely many (b) $\times 6$ gives $4x - 6y = 24$, not 18: $y = \frac{2}{3}x - 4$ and $y = \frac{2}{3}x - 3$, no solution (c) $\times 4$ gives $3x + 4y = 8$: $y = -\frac{3}{4}x + 2$ and $y = -\frac{3}{2}x + 5$, one solution (d) 3(a) ($\times 2$) and 4(a) ($\times 6$) are true multiples; multiplying an equation by a non-zero constant changes neither m nor b, so it is the same line. In 3(b) and 4(b) the coefficients were multiplied but the constant was not, which shifts b and makes the lines parallel. **5.** (a) two lines crossing at one point: one solution, $(2, 1)$ (b) two parallel lines, both slope $\frac{1}{2}$ ($b = 3$ and $b = -2$): no solution (c) the dashed line lies on the grey line — one line, so infinitely many solutions. **6.** $y = kx - 4$, $y = 2x + 1$: (a) $k = 2$ (equal slopes, $-4 \neq 1$) (b) impossible — the intercepts are fixed at -4 and 1 and can never be equal (c) every $k \neq 2$. **7.** $y = 5x + k$, $y = 5x - 3$: (a) every $k \neq -3$ (b) $k = -3$ (c) impossible — both slopes are 5 no matter what k is, so the lines are never crossing lines. **8.** $y = -\frac{k}{3}x + 3$ and $y = -2x + 3$: (a) impossible — both intercepts are 3, so equal slopes forces the same line (b) $k = 6$ (c) every $k \neq 6$. **9.** $y = -2x + 5$ and $y = -2x + k$: (a) every $k \neq 5$ (b) $k = 5$ (c) impossible — both slopes are -2. **10.** $y = \frac{12}{k}x - \frac{24}{k}$ and $y = \frac{4}{3}x - \frac{10}{3}$. (a) $k = 9$: check $y = \frac{4}{3}x - \frac{8}{3}$ vs $y = \frac{4}{3}x - \frac{10}{3}$ — same slope, different intercept, no solution (b) impossible: $k = 9$ is the only value making the slopes match, and it gives different intercepts (c) every $k \neq 9$, including $k = 0$, which turns the first equation into the vertical line $x = 2$ and still gives one crossing point. **11.** Different-looking is not the same as different. Dividing $6x - 4y = 10$ by 2 gives $3x - 2y = 5$ — the second equation exactly. Both are $y = \frac{3}{2}x - \frac{5}{2}$, so it is one line drawn twice: *infinitely many* solutions, not one. **12.** No. Two distinct points determine exactly one line, so if two lines share two points they must be the same line — and then they share every point, infinitely many. The only possible counts are 0, 1, and infinitely many. **13.** No. Different slopes means the lines tilt differently, so they must meet, and they can meet only once: exactly one solution. The y-intercepts are not needed — you only check them when the slopes are equal. **14.** $3x - 6y = 12$ is $y = \frac{1}{2}x - 2$. (a) any line with $m = \frac{1}{2}$ and $b \neq -2$, e.g. $x - 2y = 10$ ($y = \frac{1}{2}x - 5$) (b) any line with $m \neq \frac{1}{2}$, e.g. $x + y = 1$ (c) any non-zero multiple, e.g. $x - 2y = 4$ or $6x - 12y = 24$. **15.** $y = 25x + 40$ and $y = 25x + 75$: same slope, different intercepts, so *no solution.* Gym B costs exactly \$35 more than Gym A every single month, so the totals never become equal. **16.** $y = 12x + 30$ and $3y = 36x + 90$, which simplifies to $y = 12x + 30$ — the same equation, so *infinitely many* solutions. The two clerks are describing the identical price, just worded differently. **17.** (a) $L_1 : y = 2x - 1$, $L_2 : y = -x + 5$, $L_3 : y = 2x - 7$. L_1 & L_2: slopes 2 and -1, one solution. L_1 & L_3: slope 2 both, $b = -1$ vs -7, no solution (parallel). L_2 & L_3: slopes -1 and 2, one solution. (b) $L_1 \cap L_2 = (2, 3)$ and $L_2 \cap L_3 = (4, 1)$; L_1 and L_3 never meet. A common point would have to lie on both L_1 and L_3, and those two are parallel, so no such point exists — no point lies on all three.

Lesson 3 — Extra Practice

Solving Systems by Substitution, Part 1: When a Variable Is Already Isolated

Part A — Substituting When y Is Already Isolated

1. Each system already has one equation written as "$y = \ldots$". Substitute that expression into the *other* equation, solve for x, back-substitute to find y, then state the solution as an ordered pair and check it in both original equations.

(a) $y = 2x - 1$
$4x + 3y = 17$

(b) $y = 5 - 3x$
$2x + 5y = 12$

(c) $y = 4x + 9$
$3x - 2y = -13$

(d) $y = -x - 4$
$5x + 2y = -11$

2. Solve each system by substitution and check in both equations. Some answers are negative and some are fractions — leave every answer **exact** (no decimals, no rounding).

(a) $y = 4x + 1$
$2x + 3y = 10$

(b) $y = -2x + 1$
$6x + 4y = 9$

(c) $y = x + 1$
$4x - 6y = 3$

(d) $y = 2x$
$5x - 4y = 9$

3. Solve each system by substitution. Check your solution in both equations.

(a) $y = x - 7$
$2x + y = 5$

(b) $y = -4x$
$3x + 2y = -10$

(c) $y = 5 - x$
$3x - 2y = 0$

Part B — When x Is Isolated Instead

4. In every system below it is x, not y, that is already isolated. Substitute the expression into the *other* equation — it replaces x, so the equation that is left contains only y. Solve for y first, then back-substitute to find x. Check in both equations.

(a) $x = 3y - 2$
$4x + y = 5$

(b) $x = -2y + 7$
$3x - 4y = 1$

(c) $x = y + 5$
$2x + 3y = 0$

(d) $x = 4 - 5y$
$2x - 3y = 21$

5. Solve each system by substitution. Be careful when you write the ordered pair: you find y first here, but the pair is still written (x, y).

(a) $x = 2y + 3$
$5x - 2y = 7$

(b) $x = -y - 1$
$4x + 3y = -9$

(c) $x = 3y$
$2x - 5y = 2$

Part C — Parentheses and Negative Coefficients

6. When the variable you are replacing already has a coefficient in front of it, the whole expression must go inside **parentheses** before you distribute — for example, substituting $2x - 5$ into $3y$ gives $3(2x - 5)$, not $3 \cdot 2x - 5$. Write the parentheses every time, then distribute. Check each solution in both equations.

(a) $y = 2x - 5$
$3y - 4x = -9$

(b) $y = 4 - x$
$-2y + 5x = 6$

(c) $x = 1 - 2y$
$-3x + 4y = 7$

(d) $y = -3x + 2$
$5x - 2y = 18$

7. A **negative** coefficient multiplies every term inside the parentheses, so both signs change. Substitute, distribute carefully, and check. Leave any fraction answers exact.

(a) $y = 3x - 8$
$x - 5y = 12$

(b) $x = 2 - 3y$
$4x - 7y = 27$

(c) $y = -x - 2$
$6x - 4y = 2$

Part D — Both Equations Isolated, and the Two Special Cases

8. When *both* equations are already solved for the same variable, the two right-hand sides are two names for that one value, so you may set them equal to each other immediately. This is the fastest case of all — there is nothing to rearrange. Solve each system this way, then find the second variable and check.

(a) $y = 3x - 5$
$y = x + 1$

(b) $y = -2x + 7$
$y = 4x - 5$

(c) $y = 5x + 2$
$y = 2x - 4$

(d) $x = 4y - 1$
$x = y + 8$

9. In each system below the variables disappear during the substitution and you are left with a statement containing only numbers. If that statement is **true**, every point on the line works and the system has infinitely many solutions; if it is **false**, the lines are parallel and there is no solution. Substitute, state the leftover number statement, and say which case it is.

(a) $y = 2x - 3$
$4x - 2y = 6$

(b) $y = 3x + 1$
$6x - 2y = 5$

(c) $x = 5 - y$
$2x + 2y = 10$

(d) $x = 2y + 4$
$3x - 6y = 5$

10. In Lesson 2 you met these same two special cases on a graph. Explain, in your own words, what the picture looks like when substitution ends in a statement such as $6 = 6$, and what it looks like when substitution ends in a statement such as $0 = 5$. Then explain why a leftover statement like $2x = 2x$ tells you nothing at all — and what mistake produces it.

Part E — Checking and Error Analysis

11. Rohan solved the system $y = 2x - 6,\ \ 5x - 3y = 4$ and wrote the following work.

$$5x - 3 \cdot 2x - 6 = 4 \quad \Longrightarrow \quad 5x - 6x - 6 = 4 \quad \Longrightarrow \quad -x = 10 \quad \Longrightarrow \quad x = -10$$

(a) Identify the single mistake in the first line, and explain what he should have written instead.

(b) Redo the substitution correctly and find the true solution. Check it in both original equations.

12. Asked to solve $y = -3x + 11,\ \ 2x + 5y = 3$, Priya substituted correctly, solved, and wrote "the solution is 4."

(a) What did Priya forget to do, and why is a single number never a complete answer to a system?

(b) Finish her work and state the solution as an ordered pair. Check it in both equations.

13. *(Checking Practice)* Substitution is only finished once the pair has been tested in **both original** equations — not the rearranged ones.

(a) Solve $y = 4x - 9,\ \ 3x + 2y = 4$ by substitution. Then write out the check as *two separate lines of work*: one line showing the left side and right side of $y = 4x - 9$, and one line showing the left side and right side of $3x + 2y = 4$.

(b) System: $x = 3 - 2y,\ \ 4x + y = -2$.
Claimed solution: $(-1, 2)$.
Test the claim in both equations. If it fails either one, solve the system properly.

14. System: $y = 5x + 3,\ \ 2x - y = -9$. A student claims the solution is $(2, 11)$. Test the claim in both original equations, state which equation (if any) it fails, and give the correct solution.

Part F — Working Backwards, Context, and a Stretch

15. **Work backwards.** In each part you are told the solution of the system. Use it to find the missing number. *Hint:* a solution must satisfy *both* equations, so substitute the given x- and y-values into the equation that contains the unknown letter.

(a) $y = 2x + k$
$3x + y = 17$
solution $(4, 5)$; find k.

(b) $y = ax - 1$
$2x + 3y = 13$
solution $(2, 3)$; find a.

(c) $x = 3y + c$
$5x - 2y = 16$
solution $(2, -3)$; find c.

16. The system $y = mx + 4$, $x + y = 10$ has a solution whose x-coordinate is 2. Determine the y-coordinate first, then the value of m. Explain why the order matters here.

17. Define your two variables, write a system in which one equation is already isolated, then solve it by substitution.

(a) The sum of two numbers is 32. The larger number is 5 more than twice the smaller number. Determine both numbers.

(b) A drama club sold adult tickets at \$12 and student tickets at \$7. They sold three times as many student tickets as adult tickets and took in \$660. How many of each did they sell?

18. **Interpreting the answer.** A fractional answer is perfectly fine in some contexts and impossible in others.

(a) A rectangle's length is 4 cm more than twice its width, and its perimeter is 35 cm. Determine the length and the width. Is a fractional answer acceptable here? Explain.

(b) A jar holds only nickels (5¢) and dimes (10¢). There are 3 more dimes than nickels, and the total value is \$2.00. Solve the system, then explain what your answer tells you about whether such a jar can exist.

19. **Stretch — a substitution chain.** Three numbers x, y and z satisfy $z = 2y$, $y = x + 3$, and $x + y + z = 45$. Two of these are already isolated, so you can substitute twice: replace y in the equation for z to write z in terms of x, then substitute both expressions into the third equation. Determine x, y and z, and check all three equations.

Answer Key. 1. (a) $(2, 3)$ (b) $(1, 2)$ (c) $(-1, 5)$ (d) $(-1, -3)$. **2.** (a) $\left(\frac{1}{2}, 3\right)$ (b) $\left(-\frac{5}{2}, 6\right)$ (c) $\left(-\frac{9}{2}, -\frac{7}{2}\right)$ (d) $(-3, -6)$. **3.** (a) $(4, -3)$ (b) $(2, -8)$ (c) $(2, 3)$. **4.** (a) $(1, 1)$ (b) $(3, 2)$ (c) $(3, -2)$ (d) $(9, -1)$. **5.** (a) $(1, -1)$ (b) $(-6, 5)$ (c) $(6, 2)$. **6.** (a) $3(2x-5)-4x = -9$, so $(3, 1)$ (b) $-2(4-x)+5x = 6$, so $(2, 2)$ (c) $-3(1-2y)+4y = 7$, so $(-1, 1)$ (d) $5x - 2(-3x+2) = 18$, so $(2, -4)$. **7.** (a) $(2, -2)$ (b) $(5, -1)$ (c) $\left(-\frac{3}{5}, -\frac{7}{5}\right)$. **8.** (a) $3x - 5 = x + 1$, so $(3, 4)$ (b) $(2, 3)$ (c) $(-2, -8)$ (d) $4y - 1 = y + 8$, so $(11, 3)$. **9.** (a) $4x - 2(2x - 3) = 6$ gives $6 = 6$, true — infinitely many solutions (same line) (b) $6x - 2(3x + 1) = 5$ gives $-2 = 5$, false — no solution (parallel lines) (c) $2(5 - y) + 2y = 10$ gives $10 = 10$, true — infinitely many solutions (d) $3(2y + 4) - 6y = 5$ gives $12 = 5$, false — no solution. **10.** A true statement such as $6 = 6$ means the two equations describe the *same* line, so every point on it is a solution; a false statement such as $0 = 5$ means the lines are parallel and never meet, so there is no solution. A leftover statement such as $2x = 2x$ still contains the variable and is true for every x, so it proves nothing — it is what you get when you substitute back into the equation the expression came from instead of into the other one. Start over, substituting into the other equation. **11.** (a) He dropped the parentheses: -3 must multiply the whole expression $2x - 6$, so the line should read $5x - 3(2x - 6) = 4$. Multiplying only the $2x$ left the -6 unchanged (it should have become $+18$). (b) $5x - 6x + 18 = 4$, so $-x = -14$, $x = 14$ and $y = 2(14) - 6 = 22$; solution $(14, 22)$. Check: $22 = 2(14) - 6 = 22$ ✓ and $5(14) - 3(22) = 70 - 66 = 4$ ✓. (Rohan's route gave the wrong value $x = -10$.) **12.** (a) She found only x and stopped; a system has two unknowns, so the answer must be an ordered pair — she still has to back-substitute to find y. (b) $2x+5(-3x+11) = 3$ gives $-13x = -52$, $x = 4$, then $y = -3(4) + 11 = -1$; solution $(4, -1)$. Check: $-1 = -3(4) + 11$ ✓ and $2(4) + 5(-1) = 8 - 5 = 3$ ✓. **13.** (a) $3x + 2(4x - 9) = 4$ gives $11x = 22$, $x = 2$, $y = -1$; solution $(2, -1)$. Checks: $y = 4x - 9 \to$ LS $= -1$, RS $= 4(2) - 9 = -1$ ✓; $3x + 2y = 4 \to$ LS $= 3(2) + 2(-1) = 4$, RS $= 4$ ✓. (b) $(-1, 2)$ is correct: $-1 = 3 - 2(2) = -1$ ✓ and $4(-1) + 2 = -2$ ✓. **14.** $(2, 11)$ fails *both*: $y = 5x + 3$ gives RS $= 5(2) + 3 = 13 \neq 11$, and $2x - y$ gives $2(2) - 11 = -7 \neq -9$. Correct solution: $2x - (5x + 3) = -9$ gives $-3x = -6$, $x = 2$, then $y = 5(2) + 3 = 13$, that is $(2, 13)$. **15.** (a) $5 = 2(4) + k$, so $k = -3$ (b) $3 = a(2) - 1$, so $a = 2$ (c) $2 = 3(-3) + c$, so $c = 11$. **16.** From $x + y = 10$ with $x = 2$, $y = 8$; then $8 = m(2) + 4$ gives $m = 2$. The order matters because the equation containing m has two unknowns in it — you need y from the equation that has no unknown letter before you can solve for m. **17.** (a) Let s be the smaller and L the larger: $L = 2s + 5$ and $s + L = 32$; $s + 2s + 5 = 32$ gives $s = 9$, $L = 23$. (b) Let a be adult and t be student tickets: $t = 3a$ and $12a + 7t = 660$; $12a + 21a = 660$ gives $a = 20$, $t = 60$ — 20 adult and 60 student tickets. **18.** (a) $L = 2w + 4$ and $2L + 2w = 35$; $2(2w + 4) + 2w = 35$ gives $6w = 27$, $w = \frac{9}{2}$ cm $= 4.5$ cm and $L = 13$ cm. Yes — a length may be any positive measurement, so a fraction is perfectly acceptable. (b) Let n be nickels and d dimes: $d = n + 3$ and $5n + 10d = 200$; $5n + 10(n + 3) = 200$ gives $15n = 170$, $n = \frac{34}{3}$ and $d = \frac{43}{3}$. Coins come only in whole numbers, so no such jar exists — the fractional answer means the conditions are impossible, not that the algebra went wrong. **19.** $y = x + 3$, so $z = 2(x + 3) = 2x + 6$; then $x + (x + 3) + (2x + 6) = 45$ gives $4x + 9 = 45$, $x = 9$, $y = 12$, $z = 24$. Check: $z = 2(12) = 24$ ✓, $y = 9 + 3 = 12$ ✓, $9 + 12 + 24 = 45$ ✓.

Lesson 4 — Extra Practice

Solving Systems by Substitution, Part 2: Isolating a Variable First

Part A — Choosing the Variable, Then Carrying the Work

1. For each system, state which variable you would isolate and in which equation, and give the reason. If *no* coefficient is 1 or -1, say so and name the choice that keeps the fraction simplest. Do not solve.

(a) $x + 5y = 13$
$4x - 3y = 7$

(b) $3x - y = 8$
$5x + 2y = 1$

(c) $6x + 5y = 2$
$4x - 3y = 9$

(d) $7x + 2y = 15$
$x - 9y = 4$

(e) $2x - 7y = 5$
$8x + y = 11$

(f) $4x + 3y = 10$
$9x - 2y = 7$

2. Isolate the best variable (as in Question 1), then complete all five steps. Solutions here may be negative or fractional — leave fractions exact.

(a) $x + 3y = 5, \quad 4x - 5y = -14$

(b) $5x - y = -13, \quad 2x + 3y = 5$

(c) $x - 4y = -7, \quad 6x + 5y = -13$

(d) $x + 2y = 3, \quad 3x - 4y = 1$

3. In each system the variable you isolate has a coefficient of -1, so every term on the other side changes sign. Write the isolation step on its own line before substituting, then solve and check both originals.

(a) $-x + 3y = 7, \quad 2x + 5y = 8$

(b) $4x - y = -3, \quad -2x + 5y = -3$

(c) $-x - 2y = 5, \quad 3x + 7y = -16$

(d) $5x - y = 8, \quad -3x + 2y = 5$

4. In each of these systems *every* coefficient is bigger than 1, so isolating must produce a fraction. Carry it through the substitution without clearing it away, and leave any fractional answer exact.

(a) $2x + 3y = 8, \quad 5x - 4y = -3$

(b) $2x + 5y = -4, \quad 7x + 3y = 15$

(c) $4x + 6y = 5, \quad 2x - 3y = 4$

(d) Look back at (c). Which variable did you isolate, and would the other choice have been worse? Explain in one sentence.

Part B — Two Routes, Messy Coefficients, and Special Cases

5. Solve the system $3x + y = 7, \quad 2x - 5y = -1$ **twice**, once by each route below.

(a) Route 1: isolate y in the first equation, then substitute into the second.

(b) Route 2: isolate x in the second equation, then substitute into the first.

(c) Compare your two ordered pairs. What is the same, and what is different?

(d) Which route was less work, and what feature of the system made it so?

6. Each system below has fractional or decimal coefficients. Multiply each equation by a number that clears them *first*, then isolate a variable and solve.

(a) $\frac{x}{2} + \frac{y}{3} = 4, \quad x - y = 3$

(b) $0.2x + 0.5y = 2.6, \quad x - y = -1$

7. Solve each system by substitution. One of them collapses to a *false* statement and two collapse to *true* statements. State the number of solutions and explain what the collapsed statement told you.

(a) $x - 2y = 5$
$3x - 6y = 4$

(b) $2x - y = 3$
$6x - 3y = 9$

(c) $-x + 4y = 6$
$3x - 12y = -18$

Part C — Error Analysis and Working Backwards

8. Devon is solving $2x + 3y = 8, \quad x - 2y = -3$. He isolates x in the *first* equation and writes $2x = 8 - 3y$, then $x = 8 - 3y$.

(a) Name Devon's error precisely, and write the correct isolation of x from $2x + 3y = 8$.

(b) Devon carried on with $x = 8 - 3y$. Substitute his expression into $x - 2y = -3$ and find the ordered pair he ended up with.

(c) Show that Devon's ordered pair fails the equation $2x + 3y = 8$.

(d) Solve the system correctly, using the smartest isolation available.

9. Priya is solving $3x - y = 5, \quad 2x + 3y = 7$. She isolates y in the first equation to get $y = 3x - 5$, then substitutes it back into $3x - y = 5$ and obtains $3x - (3x - 5) = 5$, that is, $5 = 5$. She concludes the system has infinitely many solutions.

(a) Verify that her substitution really does give $5 = 5$, then explain why that result is guaranteed no matter what the system is — and therefore tells her nothing.

(b) Redo Step 2 correctly and solve the system. State the ordered pair and check it in both originals.

10. Work backwards. In each part the solution is given; find the missing value(s).

(a) $x + 3y = 11$ and $kx - 2y = 4$ have solution $(2, 3)$. Find k.

(b) $2x + by = 7$ and $x - y = 4$ have solution $(5, 1)$. Find b.

(c) $3x + y = a$ and $x - 2y = b$ have solution $(4, -1)$. Find a and b.

11. Consider the system $x + 2y = 6, \quad 3x + ky = c$, where k and c are unknown constants.

(a) Isolate x in the first equation and substitute into the second. Show that the result can be written as $(k-6)y = c - 18$.

(b) Use that equation to state the values of k and c that give infinitely many solutions, the values that give no solution, and the values that give exactly one solution.

Part D — Applications

12. A rectangle has length l cm and width w cm. Its perimeter is 54 cm, and three times the length minus twice the width is 31 cm.

(a) Write a system of two equations. Then simplify the perimeter equation by dividing it through by a common factor, and say how that changes your choice of variable to isolate.

(b) Solve the system and state the length and width. Check the perimeter.

13. A two-digit number has tens digit t and units digit u, so its value is $10t + u$. The digits add to 11, and reversing the digits gives a number 27 greater than the original.

(a) Write two equations, then simplify the reversal equation until it has no coefficient larger than 1.

(b) Solve for the digits and state the original number and the reversed number.

14. A lab technician mixes x mL of a 20% acid solution with y mL of a 50% acid solution to make 60 mL of a 30% acid solution. The acid equation is $0.2x + 0.5y = 0.3(60)$.

(a) Write the volume equation, then clear the decimals from the acid equation.

(b) Solve by substitution to find how much of each solution is used.

15. A cash box holds 42 coins, all nickels (5 cents each) and quarters (25 cents each), worth \$7.50 in total. Let n be the number of nickels and q the number of quarters.

(a) Write a system of two equations, using cents for the value equation.

(b) Solve by substitution and state how many of each coin there are.

16. **Stretch.** The system $x + 2y = a, \quad 3x - y = 2a$ has a constant a in place of the usual numbers. Isolate x in the first equation and substitute, then solve for x and y *in terms of* a. Check your expressions in both original equations.

17. *(Mixed Review)* Solve each system completely, choosing the most efficient variable to isolate.

(a) $x - 3y = -1$
$5x + 2y = 12$

(b) $3x + 4y = -1$
$5x - 2y = -19$

(c) $-x + 2y = 9$
$4x + 3y = -3$

Answer Key. 1. (a) x in Eq. 1 (coefficient 1) (b) y in Eq. 1 (coefficient -1) (c) no coefficient is 1 or -1, so a fraction is unavoidable; isolating y in $4x - 3y = 9$ means dividing by only 3, the smallest coefficient in the system (d) x in Eq. 2 (e) y in Eq. 2 (f) no coefficient is 1 or -1; isolating y in $9x - 2y = 7$ means dividing by only 2, the smallest coefficient in the system. **2.** (a) $x = 5 - 3y$; $(-1, 2)$ (b) $y = 5x + 13$; $(-2, 3)$ (c) $x = 4y - 7$; $(-3, 1)$ (d) $x = 3 - 2y$; $\left(\frac{7}{5}, \frac{4}{5}\right)$. **3.** (a) $x = 3y - 7$; $(-1, 2)$ (b) $y = 4x + 3$; $(-1, -1)$ (c) $x = -5 - 2y$; $(-3, -1)$ (d) $y = 5x - 8$; $(3, 7)$. **4.** (a) $x = \dfrac{8 - 3y}{2}$; $(1, 2)$ (b) $x = \dfrac{-4 - 5y}{2}$ (smallest divisor); $(3, -2)$ (c) $x = \dfrac{4 + 3y}{2}$; $\left(\frac{13}{8}, -\frac{1}{4}\right)$ (d) Isolating x in $2x - 3y = 4$ means dividing by 2; isolating from $4x + 6y = 5$ would mean dividing by 4 or 6, giving heavier fractions — so the smallest coefficient in the system is the best target. **5.** (a) $y = 7 - 3x$, so $2x - 5(7 - 3x) = -1$, $17x = 34$, $x = 2$, $y = 1$: $(2, 1)$. (b) $x = \dfrac{5y - 1}{2}$, so $3\left(\dfrac{5y - 1}{2}\right) + y = 7$, $17y = 17$, $y = 1$, $x = 2$: $(2, 1)$. (c) The ordered pair is identical; only the algebra in between differs. (d) Route 1 — the coefficient of 1 on y in $3x + y = 7$ isolates in one step and creates no fraction. **6.** (a) $\times 6$ gives $3x + 2y = 24$; with $x = y + 3$, $5y + 9 = 24$, $y = 3$, $x = 6$: $(6, 3)$. (b) $\times 10$ gives $2x + 5y = 26$; with $x = y - 1$, $7y - 2 = 26$, $y = 4$, $x = 3$: $(3, 4)$. **7.** (a) $x = 2y + 5$ gives $15 = 4$, false: *no solution* — the lines are parallel and never meet. (b) $y = 2x - 3$ gives $9 = 9$, true: *infinitely many solutions* — the second equation is $3\times$ the first, so it is the same line. (c) $x = 4y - 6$ gives $-18 = -18$, true: *infinitely many solutions* — the second equation is $-3\times$ the first. **8.** (a) He divided only the left side by 2; every term must be divided, so $x = \dfrac{8 - 3y}{2}$, i.e. $x = 4 - \frac{3}{2}y$. (b) $(8 - 3y) - 2y = -3$ gives $y = \frac{11}{5}$ and $x = \frac{7}{5}$: $\left(\frac{7}{5}, \frac{11}{5}\right)$. (c) $2\left(\frac{7}{5}\right) + 3\left(\frac{11}{5}\right) = \frac{14}{5} + \frac{33}{5} = \frac{47}{5} \neq 8$. (d) Isolate x in $x - 2y = -3$: $x = 2y - 3$, so $2(2y - 3) + 3y = 8$, $7y = 14$, $y = 2$, $x = 1$: $(1, 2)$. **9.** (a) $3x - (3x - 5) = 3x - 3x + 5 = 5$, so the statement is $5 = 5$. The expression $y = 3x - 5$ came *from* that equation, so putting it back can only rebuild the same equation — it is true for every x and carries no new information. Substitution works only when the expression goes into the *other* equation. (b) $2x + 3(3x - 5) = 7$, $11x = 22$, $x = 2$, $y = 1$: $(2, 1)$; check $3(2) - 1 = 5$ ✓ and $2(2) + 3(1) = 7$ ✓. **10.** (a) $2k - 6 = 4$, so $k = 5$ (and $2 + 9 = 11$ checks). (b) $10 + b = 7$, so $b = -3$ (and $5 - 1 = 4$ checks). (c) $a = 3(4) + (-1) = 11$ and $b = 4 - 2(-1) = 6$. **11.** (a) $x = 6 - 2y$, so $3(6 - 2y) + ky = c$, giving $18 - 6y + ky = c$ and $(k - 6)y = c - 18$. (b) Infinitely many when $k = 6$ *and* $c = 18$ (the statement becomes $0 = 0$); no solution when $k = 6$ and $c \neq 18$ (the statement becomes $0 =$ a non-zero number); exactly one solution whenever $k \neq 6$, for any value of c. **12.** (a) $2l + 2w = 54$ and $3l - 2w = 31$. Dividing the first by 2 gives $l + w = 27$, which now has a coefficient of 1, so isolate $l = 27 - w$. (b) $3(27 - w) - 2w = 31$, $81 - 5w = 31$, $w = 10$, $l = 17$; perimeter $= 2(17 + 10) = 54$ cm ✓. **13.** (a) $t + u = 11$ and $10u + t = 10t + u + 27$, which simplifies to $9u - 9t = 27$, then $u - t = 3$. (b) $u = t + 3$, so $t + (t + 3) = 11$, $t = 4$, $u = 7$; the number is 47 and the reversal is $74 = 47 + 27$ ✓. **14.** (a) $x + y = 60$; clearing the decimals ($\times 10$) gives $2x + 5y = 180$. (b) $x = 60 - y$, so $2(60 - y) + 5y = 180$, $3y = 60$, $y = 20$ and $x = 40$: 40 mL of the 20% solution and 20 mL of the 50% solution. **15.** (a) $n + q = 42$ and $5n + 25q = 750$. (b) $n = 42 - q$, so $5(42 - q) + 25q = 750$, $20q = 540$, $q = 27$ and $n = 15$: 15 nickels and 27 quarters (check $15 + 27 = 42$ and $75 + 675 = 750$ cents). **16.** $x = a - 2y$, so $3(a - 2y) - y = 2a$, giving $3a - 7y = 2a$, $7y = a$, $y = \dfrac{a}{7}$ and $x = a - \dfrac{2a}{7} = \dfrac{5a}{7}$. Check: $\dfrac{5a}{7} + \dfrac{2a}{7} = a$ ✓ and $\dfrac{15a}{7} - \dfrac{a}{7} = \dfrac{14a}{7} = 2a$ ✓. **17.** (a) $x = 3y - 1$; $(2, 1)$ (b) no coefficient is 1 or -1 — $y = \dfrac{5x + 19}{2}$ uses the smallest divisor; $(-3, 2)$ (c) $x = 2y - 9$; $(-3, 3)$.

Lesson 5 — Extra Practice

Solving Systems by Elimination, Part 1: When Coefficients Already Match

Part A — Decide First: Add or Subtract?

Before any arithmetic, ask the one question from the lesson: *does a variable's coefficients match, or are they opposite?* Same number, **same** sign $\longrightarrow$ **subtract**. Same number, **opposite** sign $\longrightarrow$ **add**. Look at both variables — sometimes it is the x's that line up, sometimes the y's.

1. State whether you would add or subtract, and name the variable that is eliminated. **Do not solve.**

(a) $7x + 2y = 19$
$7x - 5y = -9$

(b) $4x + 9y = 1$
$-4x + 3y = 11$

(c) $6x - 5y = 22$
$11x - 5y = 42$

(d) $-3x + 8y = -4$
$-3x - 2y = 16$

(e) $2x - 7y = 13$
$9x + 7y = -2$

(f) $-5x - 4y = 3$
$5x - 9y = 22$

2. These equations are not written in the same order, so nothing lines up yet. Rewrite each system with the x-terms, y-terms and constants in columns, then state add or subtract and name the variable eliminated. **Do not solve.**

(a) $3x + 2y = 7$
$2y - 3x = 1$

(b) $4x - 3y = 18$
$-3y + 7x = 27$

(c) $2y + 5x = 16$
$5x - 6y = -8$

3. Carry out the correct operation and write *only* the resulting one-variable equation. **Do not solve it.** Remember that subtracting changes the sign of every term on that side, the constant included.

(a) $9x + 4y = -5$
$9x - 6y = -35$

(b) $-8x + 5y = -12$
$8x + 3y = 44$

(c) $6x - 11y = -7$
$6x - 4y = 14$

Part B — Carrying Out the Elimination

4. In every system below a variable's coefficients **match**, so subtract. Write the subtraction line before you simplify it, and watch the right-hand side. Solve, then back-substitute.

(a) $5x - 3y = -19, \quad 5x + 7y = 11$

(b) $-4x + 9y = -21, \quad -4x + 2y = -14$

(c) $2x - 6y = 22, \quad -3x - 6y = 42$

(d) $8x + 5y = -11, \quad 8x - 3y = 13$

5. In every system below a variable's coefficients are **opposites**, so add. Some solutions are negative or fractional — leave fractions exact.

(a) $7x + 6y = 3, \quad -7x + 2y = 29$

(b) $-9x + 4y = -53, \quad 9x + 11y = 23$

(c) $6x - 5y = 39, \quad 3x + 5y = -33$

(d) $x - 8y = -2, \quad 2x + 8y = -16$

6. Now you decide. For each system state the operation and the variable eliminated, then solve. Check your solution in both original equations.

(a) $4x + y = 25, \quad 4x - 9y = 55$

(b) $5x - 7y = -47, \quad -2x + 7y = 44$

(c) $-6x - 4y = -29, \quad 10x - 4y = -5$

(d) $-3x + 2y = -6, \quad 3x - 11y = 87$

Part C — Special Cases, Errors, and Checking

7. Eliminate as usual. One system collapses to a **false** statement, one collapses to a **true** statement, and one has exactly one solution. State which is which, say what that means about the two lines, and give the solution where there is one.

(a) $4x - 6y = 10$
$4x - 6y = 3$

(b) $5x + 2y = 8$
$-5x - 2y = -8$

(c) $4x + 5y = 7$
$4x - 5y = 17$

8. **Error analysis.** Each student below chose correctly but executed badly. Find the error, explain it in one sentence, and finish the question properly.

(a) For $5x + 3y = -1$ and $5x - 8y = 76$, Dev subtracted and wrote
$5x - 5x + 3y - 8y = -1 - 76$
$-5y = -77$, so $y = \frac{77}{5}$.

(b) For $-4x + 9y = 6$ and $-4x + 5y = -2$, Mei added and wrote
$-8x + 14y = 4$,
then said she was stuck.

9. Solve the system $9x + 4y = -17$, $9x - 5y = 1$ by elimination. Then write out the **full check**: substitute your ordered pair into *each* original equation separately and show both sides agree. A pair that satisfies only one equation is not a solution.

Part D — Working Backwards and Applications

10. Each system below is missing a number. Use the information given to find it.

(a) $(3, -1)$ solves
$7x + ky = 4$
$7x - 2y = 23$.
Find k.

(b) $(1, -2)$ solves
$-5x + 3y = c$
$-5x - 4y = 3$.
Find c.

(c) Adding $mx + 5y = 27$ and
$2x - 5y = -7$ eliminates y and gives
$x = 4$. Find m, then find y.

11. A drama club sells adult tickets at a dollars each and student tickets at s dollars each. One family bought 3 adult and 5 student tickets for \$49. Another bought 3 adult and 2 student tickets for \$34.

(a) Write a system of two equations for this situation.
(b) Explain why subtracting is the natural first move here, then find a and s.
(c) Check both prices against both purchases.

12. At a canteen, 4 muffins and 3 juices cost \$17, while 4 muffins and 7 juices cost \$29.
(a) Write a system using m and j.
(b) Solve it by elimination and state the price of each item.

13. For a rectangle, twice the length plus twice the width is 46 cm, and twice the length minus twice the width is 6 cm.
(a) Write a system using l and w.
(b) Add to find l; subtract to find w. Then check.

Part E — Stretch and Reasoning

14. Elimination gives you one variable immediately. Sometimes that is the only one you need.
(a) If $11x + 6y = 40$ and $11x - 2y = 8$, find the value of y **without ever finding** x. Then find x as well, and explain why answering only the question asked saved you the ugly part.

(b) Solve $9x + 4y = 31$, $9x - 4y = 23$ by elimination. Then explain in two sentences why substitution would have been the slower route for this particular system.

15. In the system $3x + 7y = 1$, $-3x + 7y = -29$ the x-coefficients are opposites *and* the y-coefficients match, so both operations eliminate something.
(a) Add the equations and finish solving.
(b) Subtract the equations instead and finish solving.
(c) Confirm the two routes agree, and state what has to be true of a system for both operations to work.

16. Consider the system $-6x + 7y = d$ and $-6x + 7y = 15$, where d is a constant.
(a) For which value of d does the system have infinitely many solutions?

(b) What happens for every other value of d, and what do the graphs look like?

(c) Can this system ever have exactly one solution? Explain.

17. **Mixed review.** Line the equations up first, then decide, eliminate and solve.

(a) $y + 6x = 13, \quad 6x - 4y = 8$

(b) $-8x + 3y = -1, \quad 3y + 8x = 31$

(c) $10x - 7y = -4, \quad 10x + 2y = 14$

(d) $2a + 3b = 4, \quad 2a - 3b = 28$

Answer Key. 1. (a) subtract, x (b) add, x (c) subtract, y (d) subtract, x (e) add, y (f) add, x. **2.** (a) $3x + 2y = 7$ and $-3x + 2y = 1$: the x's are opposites, so *adding* removes x; the y's also match, so *subtracting* removes y — either works. (b) $4x - 3y = 18$ and $7x - 3y = 27$: subtract, y. (c) $5x + 2y = 16$ and $5x - 6y = -8$: subtract, x (and note $16 - (-8) = 24$). **3.** (a) subtract: $10y = -5 - (-35) = 30$ (b) add: $8y = 32$ (c) subtract: $-7y = -7 - 14 = -21$. **4.** (a) $(5x - 3y) - (5x + 7y) = -19 - 11$ gives $-10y = -30$, $y = 3$, $x = -2$: $(-2, 3)$. (b) $(-4x + 9y) - (-4x + 2y) = -21 - (-14)$ gives $7y = -7$, $y = -1$, $x = 3$: $(3, -1)$. (c) $(2x - 6y) - (-3x - 6y) = 22 - 42$ gives $5x = -20$, $x = -4$, $y = -5$: $(-4, -5)$. (d) $8y = -11 - 13 = -24$, $y = -3$, $8x = 4$: $\left(\frac{1}{2}, -3\right)$. **5.** (a) $8y = 32$, $(-3, 4)$ (b) $15y = -30$, $(5, -2)$ (c) $9x = 6$, $\left(\frac{2}{3}, -7\right)$ (d) $3x = -18$, $\left(-6, -\frac{1}{2}\right)$. **6.** (a) subtract, x; $10y = -30$, $(7, -3)$ (b) add, y; $3x = -3$, $(-1, 6)$ (c) subtract, y; $-16x = -29 - (-5) = -24$, $\left(\frac{3}{2}, 5\right)$ (d) add, x; $-9y = 81$, $(-4, -9)$. **7.** (a) Subtracting gives $0 = 7$, false — **no solution**; the lines are parallel and never meet. (b) Adding gives $0 = 0$, true — **infinitely many solutions**; the two equations describe the same line. (c) Adding gives $8x = 24$, $x = 3$, $y = -1$ — one solution, $(3, -1)$; the lines cross once. **8.** (a) Dev flipped the sign of the $5x$ but not of the $-8y$: subtracting $5x - 8y$ means $-5x + 8y$, so the y-column is $3y + 8y = 11y$, not $3y - 8y$. Correctly $(5x + 3y) - (5x - 8y) = -1 - 76$ gives $11y = -77$, $y = -7$, then $5x - 21 = -1$, $x = 4$: $(4, -7)$. (b) The x-coefficients are both -4 — the *same* sign, not opposite — so adding doubles them instead of cancelling; nothing was eliminated. Subtracting gives $4y = 6 - (-2) = 8$, $y = 2$, then $-4x + 18 = 6$, $x = 3$: $(3, 2)$. **9.** Subtract: $9y = -17 - 1 = -18$, so $y = -2$; then $9x + 4(-2) = -17$, $9x = -9$, $x = -1$. Solution $(-1, -2)$. Check Eq. 1: $9(-1) + 4(-2) = -9 - 8 = -17$ ✓. Check Eq. 2: $9(-1) - 5(-2) = -9 + 10 = 1$ ✓. **10.** (a) $7(3) + k(-1) = 4$, so $21 - k = 4$ and $k = 17$. (b) $c = -5(1) + 3(-2) = -11$. (c) Adding gives $(m + 2)x = 20$; with $x = 4$, $m + 2 = 5$, so $m = 3$, and $2(4) - 5y = -7$ gives $y = 3$. **11.** (a) $3a + 5s = 49$, $3a + 2s = 34$. (b) The a-coefficients are both 3 — same number, same sign — so subtracting removes a immediately: $(3a + 5s) - (3a + 2s) = 49 - 34$ gives $3s = 15$, $s = 5$; then $3a + 2(5) = 34$, $3a = 24$, $a = 8$. Adult \$8, student \$5. (c) $3(8) + 5(5) = 24 + 25 = 49$ ✓ and $3(8) + 2(5) = 24 + 10 = 34$ ✓. **12.** (a) $4m + 3j = 17$, $4m + 7j = 29$. (b) Subtract: $-4j = -12$, $j = 3$; then $4m + 9 = 17$, $m = 2$. Muffins \$2, juice \$3. **13.** (a) $2l + 2w = 46$, $2l - 2w = 6$. (b) Adding: $4l = 52$, $l = 13$; subtracting: $4w = 40$, $w = 10$. Check: $26 + 20 = 46$ ✓ and $26 - 20 = 6$ ✓. **14.** (a) Subtract: $8y = 40 - 8 = 32$, so $y = 4$ in one step. Then $11x = 40 - 24 = 16$, $x = \frac{16}{11}$ — the fraction only appears if you actually need x. (b) Add: $18x = 54$, $x = 3$; then $4y = 31 - 27 = 4$, $y = 1$: $(3, 1)$. Substitution would force $y = \frac{31 - 9x}{4}$, dragging a fraction through every later step, while adding cleared y in one line. **15.** (a) Adding: $14y = 1 + (-29) = -28$, $y = -2$; then $3x - 14 = 1$, $x = 5$. (b) Subtracting: $6x = 1 - (-29) = 30$, $x = 5$; then $15 + 7y = 1$, $y = -2$. (c) Both give $(5, -2)$. Both operations work exactly when one variable's coefficients match *and* the other variable's coefficients are opposites. **16.** (a) $d = 15$ — the two equations become identical, subtracting gives $0 = 0$, and every point on the line $-6x + 7y = 15$ is a solution. (b) For any $d \neq 15$, subtracting gives $0 = d - 15$, a false statement: no solution. The graphs are two distinct parallel lines (both with slope $\frac{6}{7}$). (c) No. The left sides are identical, so the lines have the same slope — they are either the same line or parallel, and parallel lines never cross exactly once. **17.** (a) $6x + y = 13$ and $6x - 4y = 8$: subtract, $5y = 13 - 8 = 5$, $y = 1$, $x = 2$: $(2, 1)$. (b) $-8x + 3y = -1$ and $8x + 3y = 31$: add, $6y = 30$, $y = 5$, $x = 2$: $(2, 5)$. (c) subtract, $-9y = -4 - 14 = -18$, $y = 2$, $x = 1$: $(1, 2)$. (d) add, $4a = 32$, $a = 8$, $3b = 4 - 16 = -12$, $b = -4$: $a = 8$, $b = -4$.

Lesson 6 — Extra Practice

Solving Systems by Elimination, Part 2: Multiplying First

Part A — Plan Before You Multiply

1. For each system, state which equation (or equations) you would multiply, by what constant, and which variable will vanish. Say whether the job needs *one* equation or *both*. Do not solve.

(a) $3x + 2y = 12$
$x - 5y = 4$
(eliminate x)

(b) $4x - 3y = 7$
$6x + 5y = 2$
(eliminate x)

(c) $5x + 4y = 1$
$3x - 2y = 8$
(eliminate y)

(d) $2x + 9y = 5$
$7x - 6y = 3$
(eliminate y)

(e) $6x - 5y = 11$
$4x + 3y = 9$
(eliminate x)

(f) $x - 7y = 2$
$8x + 3y = 5$
(eliminate x)

2. For each system, find the lowest common multiple of the x-coefficients *and* the lowest common multiple of the y-coefficients. State which variable is **cheaper** to eliminate and give your reason. Do not solve.

(a) $4x + 3y = 10$
$6x - 5y = 2$

(b) $2x + 7y = 12$
$6x - 5y = 8$

(c) $4x + 6y = 22$
$9x - 4y = 5$

3. In each system below the two coefficients of the variable named have the *same* sign, so matching them would force you to subtract. Give a **negative** multiplier instead, so the coefficients become opposites and the equations can simply be added. State the multiplier and the equation it applies to. Do not solve.

(a) $3x + 2y = 11$
$6x + 5y = 26$
(eliminate x)

(b) $2x - 7y = 4$
$8x - 3y = 5$
(eliminate x)

(c) $9x + 4y = 6$
$3x + 10y = 1$
(eliminate x)

Part B — Multiplying One Equation

4. In each system one coefficient is already a factor of the other, so only *one* equation needs multiplying. Multiply **every term, including the constant**, then eliminate, solve, and verify the pair in both originals. Leave any fraction exact.

(a) $x + 4y = -1, \quad 3x - 5y = 14$

(b) $2x - y = 7, \quad 5x + 3y = 1$

(c) $3x + 2y = 5, \quad 9x - 5y = -29$

(d) $4x + y = 3, \quad 6x - 5y = -2$

Part C — Multiplying Both Equations

5. In each system neither coefficient divides the other, so **both** equations must be multiplied. Use the *lowest* common multiple so the numbers stay as small as possible. Solve and verify; leave any fraction exact.

(a) $3x + 4y = 6, \quad 5x - 6y = -28$

(b) $4x - 3y = 14, \quad 6x + 5y = 2$

(c) $5x + 2y = -5, \quad 3x - 7y = -44$

(d) $8x + 6y = 9, \quad 6x - 4y = 11$

Part D — Aligning and Clearing First

6. These equations are not yet in the arrangement elimination needs. Rewrite each system with the variable terms on the left in the same order and the constant on the right, then choose your multipliers and solve.

(a) $3x = 14 - 2y, \quad 5y = 4x - 11$

(b) $2y - 5x = -16, \quad 7x = 3y + 23$

(c) $4x + 9 = 3y, \quad 2x - 5y = -1$

7. Clear the fractions or decimals *first* by multiplying each equation through by a suitable number — every term, both sides — then eliminate as usual. Leave any fractional answer exact.

(a) $\dfrac{x}{2}+\dfrac{y}{3}=4, \quad \dfrac{x}{4}-\dfrac{y}{2}=-1$

(b) $0.2x+0.5y=2.9, \quad 0.3x-0.4y=-1.4$

(c) $\dfrac{x+y}{2}=5, \quad \dfrac{x-y}{3}=1$

Part E — Errors, Special Cases, and Missing Numbers

8. Devon solved $2x+3y=12$ and $5x-y=13$. He decided to eliminate y by multiplying Equation 2 by 3, and wrote $15x-3y=13$. Adding gave him $17x=25$, so $x=\frac{25}{17}$.
(a) Identify his error in one sentence.
(b) Write the correct multiplied Equation 2 and finish the system properly.

9. Priya solved $3x+4y=2$ and $5x-6y=16$. She multiplied Equation 1 by 3 and Equation 2 by 2, getting $9x+12y=6$ and $10x-12y=32$. She then *subtracted* and obtained $-x+24y=-26$.
(a) Her multiplying was correct. Explain why subtracting was the wrong move here.
(b) Redo the last step correctly and solve the system.

10–11. Multiply as planned, then interpret the statement you are left with. In Lesson 2 you saw that two equations can describe the *same line in disguise*; connect each result below to that idea.

10. Eliminate x from $6x-4y=10$ and $9x-6y=21$. State what happens, what it tells you about the graphs, and how many solutions the system has.

11. Eliminate x from $4x+6y=10$ and $6x+9y=15$. State what happens, what it tells you about the graphs, and how many solutions the system has. Give two ordered pairs that both satisfy the system.

12. **Work backwards.** In each case the solution — or the number of solutions — is given. Find the missing coefficient.

(a) $5x + ky = 1$ and $2x - 3y = 12$ have the solution $(3, -2)$. Find k.

(b) $ax - 4y = 10$ and $3x + 5y = 21$ have the solution $(2, 3)$. Find a.

(c) $mx + 6y = 9$ and $4x + 3y = 5$ have *no* solution. Find m, and show the elimination that proves it.

Part F — Applications

13. A lab technician mixes a 20% acid solution with a 50% acid solution to obtain 300 mL of a 30% acid solution. Let x be the millilitres of the 20% solution and y the millilitres of the 50% solution.

(a) Write a system of two equations: one for the total volume, one for the acid content.

(b) Clear the decimals, then solve by elimination. State how much of each solution is used and verify both equations.

14. Maya trains at two steady speeds. On Monday she cycled for 2 h and ran for 3 h, covering 46 km in total. On Thursday she cycled for 4 h and ran for 1 h, covering 62 km in total.

(a) Let c be her cycling speed and r her running speed, both in km/h. Write a system of two equations.

(b) Solve by elimination to find both speeds.

15. A theatre sold 5 adult tickets and 3 student tickets for a total of \$97, and on another night sold 4 adult tickets and 7 student tickets for a total of \$119.

(a) Let a be the price of an adult ticket and s the price of a student ticket. Write a system of two equations.

(b) Solve by elimination to find both prices.

16. At an arcade, 3 red tokens and 7 blue tokens are worth 145 points, while 5 red tokens and 4 blue tokens are worth 165 points.

(a) Let r be the point value of a red token and b that of a blue token. Write a system of two equations.

(b) Decide which variable is cheaper to eliminate, justify the choice, then solve and verify.

Part G — Stretch

17. **Literal coefficients.** Solve the system $2x + 3y = a$ and $5x - 2y = 4a$ for x and y in terms of the constant a. Multiply exactly as you would with numbers, and verify your answers by substituting them into both original equations.

18. **Two routes, one system.** Consider $4x + 5y = 7$ and $6x - 3y = 21$.

(a) Solve the system by elimination, multiplying whichever equation or equations you judge cheapest. Show every step.

(b) Solve the same system by substitution. *Hint:* divide Equation 2 by 3 first — that produces a coefficient of 1 to isolate.

(c) The two answers must agree. Which method took fewer steps here, and what feature of Equation 2 made it so? Describe the kind of system for which your answer would flip.

Answer Key. 1. (a) Eq. 2 by 3, then subtract; one equation. (b) Eq. 1 by 3 and Eq. 2 by 2 (LCM 12), then subtract; both. (c) Eq. 2 by 2, then add; one equation. (d) Eq. 1 by 2 and Eq. 2 by 3 (LCM 18), then add; both. (e) Eq. 1 by 2 and Eq. 2 by 3 (LCM 12), then subtract; both. (f) Eq. 1 by 8, then subtract; one equation. **2.** (a) x: LCM$(4,6) = 12$; y: LCM$(3,5) = 15$. Eliminate x — smaller LCM, so smaller multipliers (3 and 2). (b) x: LCM$(2,6) = 6$; y: LCM$(7,5) = 35$. Eliminate x — and because 6 is a multiple of 2, only *one* equation needs multiplying. (c) x: LCM$(4,9) = 36$; y: LCM$(6,4) = 12$. Eliminate y. (Dividing Eq. 1 by 2 first gives $2x + 3y = 11$ and lowers the x-LCM to 18, but y is still cheaper.) **3.** (a) Eq. 1 by -2 (gives $-6x - 4y = -22$), then add. (b) Eq. 1 by -4, then add. (c) Eq. 2 by -3, then add. **4.** (a) Eq. 1 by 3, subtract: $17y = -17$; $(3,-1)$. (b) Eq. 1 by 3, add: $11x = 22$; $(2,-3)$. (c) Eq. 1 by 3, subtract: $11y = 44$; $(-1,4)$. (d) Eq. 1 by 5, add: $26x = 13$; $\left(\frac{1}{2},1\right)$. **5.** (a) Eq. 1 by 3, Eq. 2 by 2 (LCM 12 on y), add: $19x = -38$; $(-2,3)$. (b) Eq. 1 by 5, Eq. 2 by 3 (LCM 15 on y), add: $38x = 76$; $(2,-2)$. (c) Eq. 1 by 7, Eq. 2 by 2 (LCM 14 on y), add: $41x = -123$; $(-3,5)$. (d) Eq. 1 by 2, Eq. 2 by 3 (LCM 12 on y), add: $34x = 51$; $\left(\frac{3}{2},-\frac{1}{2}\right)$. **6.** (a) $3x + 2y = 14$ and $-4x + 5y = -11$; $(4,1)$. (b) $-5x + 2y = -16$ and $7x - 3y = 23$; $(2,-3)$. (c) $4x - 3y = -9$ and $2x - 5y = -1$; $(-3,-1)$. **7.** (a) $\times 6$ and $\times 4$: $3x + 2y = 24$ and $x - 2y = -4$; add: $4x = 20$, so $\left(5,\frac{9}{2}\right)$. (b) $\times 10$ both: $2x + 5y = 29$ and $3x - 4y = -14$; $(2,5)$. (c) $\times 2$ and $\times 3$: $x + y = 10$ and $x - y = 3$; $\left(\frac{13}{2},\frac{7}{2}\right)$. **8.** (a) He multiplied the left side by 3 but left the constant as 13 — every term, including the constant, must be multiplied. (b) $15x - 3y = 39$; adding $2x + 3y = 12$ gives $17x = 51$, so $x = 3$ and $y = 2$: $(3,2)$. Check $2(3) + 3(2) = 12$ ✓ and $5(3) - 2 = 13$ ✓. **9.** (a) The y-terms were already *opposites* ($+12y$ and $-12y$); subtracting turns $-12y$ into $+12y$, so the y-terms reinforce instead of cancelling — her result still contains both variables. Subtract only when the coefficients match in sign; add when they are opposites. (b) Adding: $19x = 38$, $x = 2$, then $3(2) + 4y = 2$ gives $y = -1$: $(2,-1)$. Check $5(2) - 6(-1) = 16$ ✓. **10.** Eq. 1 by 3 and Eq. 2 by 2 give $18x - 12y = 30$ and $18x - 12y = 42$; subtracting leaves $0 = -12$, which is false. The left sides are identical but the constants differ, so the lines are parallel and distinct — *no solution.* **11.** Eq. 1 by 3 and Eq. 2 by 2 both give $12x + 18y = 30$; subtracting leaves $0 = 0$, which is always true. Both equations reduce to $2x + 3y = 5$ — the same line in disguise — so there are *infinitely many* solutions. E.g. $(1,1)$ and $(4,-1)$. **12.** (a) $5(3) + k(-2) = 1$, so $15 - 2k = 1$ and $k = 7$ (Eq. 2 checks: $6 + 6 = 12$). (b) $a(2) - 4(3) = 10$, so $2a = 22$ and $a = 11$ (Eq. 2 checks: $6 + 15 = 21$). (c) $m = 8$: Eq. 2 by 2 gives $8x + 6y = 10$ versus $8x + 6y = 9$, and subtracting leaves $0 = 1$, a false statement — parallel lines, no solution. **13.** (a) $x + y = 300$ and $0.2x + 0.5y = 90$. (b) $\times 10$: $2x + 5y = 900$; Eq. 1 by 2 gives $2x + 2y = 600$; subtracting: $3y = 300$, $y = 100$ and $x = 200$. So 200 mL of the 20% solution and 100 mL of the 50% solution. Check: $200 + 100 = 300$ ✓ and $40 + 50 = 90$ mL of acid ✓. **14.** (a) $2c + 3r = 46$ and $4c + r = 62$. (b) Eq. 1 by 2, subtract (or Eq. 2 by 3 and subtract): $c = 14$ km/h cycling and $r = 6$ km/h running. Check $28 + 18 = 46$ ✓, $56 + 6 = 62$ ✓. **15.** (a) $5a + 3s = 97$ and $4a + 7s = 119$. (b) Eq. 1 by 4 and Eq. 2 by 5 (LCM 20 on a), subtract: $23s = 207$, $s = 9$ and $a = 14$. Adult \$14, student \$9. **16.** (a) $3r + 7b = 145$ and $5r + 4b = 165$. (b) r: LCM$(3,5) = 15$; b: LCM$(7,4) = 28$ — r is cheaper. Eq. 1 by 5 and Eq. 2 by 3, subtract: $23b = 230$, $b = 10$ and $r = 25$. Check $75 + 70 = 145$ ✓ and $125 + 40 = 165$ ✓. **17.** Eq. 1 by 2 and Eq. 2 by 3 (LCM 6 on y): $4x + 6y = 2a$ and $15x - 6y = 12a$; adding gives $19x = 14a$, so $x = \dfrac{14a}{19}$ and $y = \dfrac{a - 2x}{3} = \dfrac{-3a}{19}$. Check: $\dfrac{28a}{19} - \dfrac{9a}{19} = a$ ✓ and $\dfrac{70a}{19} + \dfrac{6a}{19} = 4a$ ✓. **18.** (a) Eq. 2 by 5 and Eq. 1 by 3 (LCM 15 on y) — or, cheaper, divide Eq. 2 by 3 to get $2x - y = 7$, multiply that by 5 and add to Eq. 1: $14x = 42$, $x = 3$, $y = -1$: $(3,-1)$. (b) From $2x - y = 7$, $y = 2x - 7$; then $4x + 5(2x - 7) = 7$ gives $14x = 42$, $x = 3$ and $y = -1$ — same answer. (c) Substitution was at least as quick because Equation 2 divides by 3 into $2x - y = 7$, whose y-coefficient is -1, so isolating creates no fraction. Had every coefficient been larger than 1 with no common factor (say $6x - 7y = 21$), isolating would force a fraction and elimination would win.

Lesson 7 — Extra Practice

Solving Word Problems with Systems of Equations

Part A — Translating Only: Define, Then Write Two Equations

1. For each situation, define the two variables (say exactly what each letter counts, with units) and write a system of two equations that models it. **Do not solve.**

(a) A food truck at the Calgary Stampede sells poutine and mini-donuts. On Saturday it sold 3 times as many orders of mini-donuts as orders of poutine, 96 orders in all.

Let _____

Let _____

(b) A cash box at a Lethbridge car wash holds only loonies and toonies. There are 53 coins, worth \$81 altogether.

Let _____

Let _____

(c) A rectangular plot in an Edmonton community garden is 4 m longer than it is wide. Its perimeter is 68 m.

Let _____

Let _____

(d) Two service clubs collected 240 kg of food for the Red Deer food bank. The senior club collected 30 kg more than twice what the junior club collected.

Let _____

Let _____

2. These translations are harder — the two facts are buried deeper, and some use ideas beyond the lesson's examples (percent, speed, digits). Define the variables and write the system. **Do not solve.**

(a) A canoe travels 24 km down the Bow River in 2 h, and the same 24 km back upstream in 3 h. *Hint:* going downstream the current adds to the paddling speed; going upstream it subtracts.

(b) A feed supplier blends a 12% protein ration with a 20% protein ration to make 400 kg of a 17% protein ration.

(c) The digits of a two-digit number add to 12. Reversing the digits produces a number that is 18 greater than the original. *Hint:* a number with tens digit t and units digit u has value $10t + u$.

(d) A hall in Grande Prairie charges a \$250 booking fee plus \$18 per guest. A second hall charges a \$400 fee plus \$13 per guest. Write the two equations whose intersection gives the number of guests at which the two halls cost the same.

3. *(Multiple Choice)* A skating rink sells adult passes for \$9 and youth passes for \$6. It sold 25 more youth passes than adult passes and took in \$1230. If a is the number of adult passes and y is the number of youth passes, which system models this situation?

A. $a + y = 25,\ 9a + 6y = 1230$ B. $a = y + 25,\ 9a + 6y = 1230$ C. $y = a + 25,\ 6a + 9y = 1230$ D. $y = a + 25,\ 9a + 6y = 1230$

Part B — Numbers, Ages, and Relationships

4. Define the variables, write a system, solve it, and answer in a sentence.

(a) The sum of two numbers is 128 and their difference is 34. Find the two numbers.

(b) One number is 3 more than twice the other. Their sum is 72. Find the two numbers.

5. The sum of two numbers is 57. When the larger number is divided by the smaller, the quotient is 2 and the remainder is 6 — that is, the larger is 6 more than twice the smaller. Find both numbers, then check your answer in the original words.

Answer: _____

6. Three years ago, Maya was exactly twice as old as her cousin Owen. Today the sum of their ages is 39. Find both of their ages today. *Hint:* let m be Maya's age today and n be Owen's age today, so three years ago they were $m-3$ and $n-3$.

Answer: _____

Part C — Two Prices: Tickets, Food, and Coins

7. The Vermilion high school drama club sold 412 tickets to its spring production. Adult tickets cost \$14 and student tickets cost \$8, and ticket sales totalled \$4142. How many tickets of each kind were sold?

Answer: _____

8. A coffee stand in Canmore sells a small coffee for \$2.75 and a large coffee for \$4.25. On Friday it sold 180 coffees for a total of \$633.00. How many of each size were sold? *Hint:* multiply the value equation by 4 (or by 100) to clear the decimals before you eliminate.

Answer: _____

9. A change machine at a Fort McMurray rec centre is emptied and found to contain only quarters and loonies: 156 coins with a total value of \$96.75. Notice that one equation counts *how many* coins there are and the other totals their *value* — these are two genuinely different facts. How many coins of each kind are there?

Answer: _____

10. **Does the answer make sense?** A charity barbecue in Airdrie sells burgers at \$6 and hot dogs at \$4. The organizers report that 95 items were sold and \$497 was collected.
(a) Write and solve the system.
(b) The solution is mathematically correct, but it cannot be the answer to this problem. Explain why, and state what that tells you about the organizers' report.

Answer: _____

Part D — Mixture and Percent Problems (percent mixtures go beyond the lesson: Stretch)

11. A lab technician needs 500 mL of a 32% antifreeze mixture. She has a 20% stock solution and a 50% stock solution on hand. How many millilitres of each should she use? *Hint:* the second equation totals the *pure antifreeze*, so the right side is $0.32(500)$. Multiply through by 100 to clear the decimals.

Answer: _____

12. A feedlot near Brooks mixes a 14% protein barley ration with a 26% protein canola ration to produce 900 kg of an 18% protein ration. How many kilograms of each ration are used? Check that the protein in your two amounts really does add up to 18% of 900 kg.

Answer: _____

Part E — Distance, Rate, and Time (Stretch: beyond the lesson's examples)

13. Priya drove a 750 km trip from Calgary to Fort McMurray. Part of the trip she drove at 100 km/h, and part of it, through a construction zone, at 60 km/h. The drive took 8.5 h of driving time. How many hours did she spend at each speed? *Hint:* define each variable as a number of *hours*; one equation totals the time and the other totals the distance, using $d = rt$.

Answer: _____

14. A jet boat travels 60 km down the Athabasca River in 2 h. The return trip of 60 km against the current takes 3 h. Let b be the boat's speed in still water and c be the speed of the current, both in km/h.
(a) Explain in one sentence why the downstream speed is $b + c$ and the upstream speed is $b - c$.
(b) Write and solve a system to find b and c.

Answer: _____

Part F — Geometry, Break-Even, and Rates of Work (Stretch: beyond the lesson's examples)

15. A rectangular skating rink in Okotoks has a perimeter of 174 m. Its length is 3 m less than twice its width. Find the length and the width, then state the area of the rink.

Answer: _____

16. **Break-even.** A Red Deer climbing gym offers two options. Under the drop-in plan you pay \$18 per visit and nothing else. Under the membership plan you pay \$120 once, plus \$6 per visit. Let n be the number of visits in a year and C be the total yearly cost in dollars.
(a) Write an equation for each plan, then solve the system to find the number of visits at which the two plans cost the same, and that common cost.
(b) Which plan is cheaper for a climber who visits 15 times in a year? Show the cost of each.
(c) If you graphed both equations, what feature of the graph is your answer to part (a)?

Answer: _____

17. Tia and Ben plant seedlings at a Sherwood Park garden centre, each at a constant hourly rate. On Monday, Tia worked 5 h and Ben worked 3 h, and together they planted 340 seedlings. On Tuesday, Tia worked 2 h and Ben worked 6 h, and together they planted 280 seedlings. Find each person's hourly planting rate.

Answer: _____

Part G — Error Analysis, Checking, and Working Backwards

18. **Error analysis: one useless equation.** A theatre in Medicine Hat charges \$5 for a small popcorn and \$8 for a large. One evening it sold 60 more smalls than larges and took in \$1080. Let s be the number of smalls and ℓ be the number of larges. Marcus wrote:

$$s - \ell = 60 \qquad \text{and} \qquad s = \ell + 60.$$

(a) Show what happens when Marcus tries to solve his system, and explain what has gone wrong with his *pair* of equations.
(b) Which fact from the problem did Marcus never use? Write a correct system and solve it.

Answer: _____

19. **Error analysis: reversed quantities.** A community theatre in Wetaskiwin sold 120 tickets: adults at \$15 and youth at \$10. The revenue from adult tickets was \$1300 *more than* the revenue from youth tickets. Nadia let a be the number of adult tickets and y the number of youth tickets, and wrote:

$$a + y = 120 \qquad \text{and} \qquad 10y - 15a = 1300.$$

(a) Solve Nadia's system. What does her answer say about the number of adult tickets, and why is that impossible?
(b) Explain exactly which quantity she reversed, write the corrected second equation, and solve the corrected system.

Answer: _____

20. **Checking in the original words.** An outfitter in Grande Cache rents canoes at \$45 per day and kayaks at \$30 per day. On the long weekend, 26 boats were rented for a total of \$990. Jonah says the answer is 16 canoes and 10 kayaks.
(a) Check Jonah's answer against *each sentence of the problem*, not against a system of equations. State which sentence his answer satisfies and which one it fails, with the arithmetic.
(b) Write and solve a correct system, then check your own answer in the original words the same way.

Answer: _____

21. **Working backwards.** In each case a number has been erased from the problem, but the answer is known. Reconstruct the missing number.

(a) "A rink sold youth passes at \$6 and adult passes at \$11. A total of 140 passes were sold for \$_____." The answer key reads: 85 youth passes and 55 adult passes. Find the erased total.

(b) "A 40 kg coffee blend mixes dark roast at \$22/kg with house roast at \$_____/kg, and the whole blend costs \$680." The answer key reads: 15 kg of dark roast. Find the erased price.

Part H — Stretch

22. **One number you do not need.** A Grade 10 class in Leduc ran a two-day bottle drive. They receive \$0.10 for each pop can and \$0.25 for each large juice bottle. On Day 1 they collected 620 containers. Over the two days they collected 1500 containers altogether — cans and bottles only — worth \$255.
(a) Which number given in the problem is *not* needed to answer part (b)? Explain how you can tell.
(b) How many cans and how many bottles were collected over the two days? Check your answer in the original words.

Answer: _____

23. **Two stages.** Tickets to a Rocky Mountain House rodeo cost \$18 for adults and \$9 for youth. On Friday, 240 tickets were sold for \$3384. On Saturday, the number of adult tickets sold was double Friday's, while the number of youth tickets sold was the same as Friday's.
(a) Write and solve a system to find how many adult and how many youth tickets were sold on Friday.
(b) Use your answer to part (a) to find Saturday's ticket revenue. *Note:* part (b) needs no new system — once the Friday counts are known, Saturday is arithmetic.
(c) State in a sentence how many tickets in total were sold over the two days.

Answer: _____

Answer Key. 1. (a) Let p = orders of poutine, d = orders of mini-donuts: $d = 3p$, $p + d = 96$. (b) Let L = number of loonies, T = number of toonies: $L + T = 53$, $1L + 2T = 81$. (c) Let l = length in m, w = width in m: $l = w + 4$, $2l + 2w = 68$. (d) Let s = kg from the senior club, j = kg from the junior club: $s + j = 240$, $s = 2j + 30$. **2.** (a) Let b = canoe speed in still water (km/h), c = current speed (km/h): $2(b + c) = 24$, $3(b - c) = 24$. (b) Let x = kg of the 12% ration, y = kg of the 20% ration: $x + y = 400$, $0.12x + 0.20y = 0.17(400) = 68$. (c) Let t = tens digit, u = units digit: $t + u = 12$, $10u + t = 10t + u + 18$ (which tidies to $u - t = 2$). (d) Let n = number of guests, C = total cost in dollars: $C = 250 + 18n$, $C = 400 + 13n$. **3.** D. There are 25 *more youth* than adults, so $y = a + 25$ (not $a = y + 25$, and not $a + y = 25$), and each adult pass is \$9, so the \$9 must multiply a. **4.** (a) $x + y = 128$, $x - y = 34$; adding gives $2x = 162$, $x = 81$, $y = 47$. The numbers are 81 and 47 ($81 + 47 = 128$ ✓, $81 - 47 = 34$ ✓). (b) $x = 2y + 3$, $x + y = 72$; $3y + 3 = 72$, $y = 23$, $x = 49$. The numbers are 49 and 23 ($49 + 23 = 72$ ✓, $2(23) + 3 = 49$ ✓). **5.** $L + S = 57$, $L = 2S + 6$; $3S + 6 = 57$, $S = 17$, $L = 40$. The numbers are 40 and 17. In words: $40 + 17 = 57$ ✓, and $40 \div 17$ is 2 with remainder 6 ✓. **6.** $m - 3 = 2(n - 3)$, so $m = 2n - 3$; with $m + n = 39$, $3n - 3 = 39$, $n = 14$ and $m = 25$. Maya is 25 and Owen is 14. Check in words: three years ago they were 22 and 11, and $22 = 2 \times 11$ ✓; $25 + 14 = 39$ ✓. **7.** $a + s = 412$, $14a + 8s = 4142$. Substituting $s = 412 - a$: $6a + 3296 = 4142$, $a = 141$, $s = 271$. They sold 141 adult tickets and 271 student tickets ($14 \times 141 = \$1974$, $8 \times 271 = \$2168$, total \$4142 ✓). **8.** $s + l = 180$, $2.75s + 4.25l = 633$. Multiplying the second by 4: $11s + 17l = 2532$; substituting $s = 180 - l$ gives $1980 + 6l = 2532$, $l = 92$, $s = 88$. They sold 88 small and 92 large ($\$242 + \$391 = \$633$ ✓). **9.** $q + n = 156$ (count) and $0.25q + 1.00n = 96.75$ (value). Multiplying the value equation by 4: $q + 4n = 387$; subtracting the count equation, $3n = 231$, $n = 77$ and $q = 79$. There are 79 quarters and 77 loonies ($\$19.75 + \$77.00 = \$96.75$ ✓). **10.** (a) $b + h = 95$, $6b + 4h = 497$; $2b + 380 = 497$, so $b = 58.5$ and $h = 36.5$. (b) You cannot sell half a burger, so no whole number of burgers and hot dogs fits both facts — the organizers' numbers must be wrong somewhere (a miscount, or money from something other than burgers and hot dogs). **11.** Let x = mL of 20% and y = mL of 50%: $x + y = 500$ and $0.20x + 0.50y = 0.32(500) = 160$, i.e. $2x + 5y = 1600$. Substituting $x = 500 - y$: $3y = 600$, $y = 200$ and $x = 300$. Use 300 mL of the 20% solution and 200 mL of the 50% solution ($60 + 100 = 160$ mL of pure antifreeze, and $160 \div 500 = 32\%$ ✓). **12.** $x + y = 900$ and $0.14x + 0.26y = 0.18(900) = 162$, i.e. $7x + 13y = 8100$. Substituting $x = 900 - y$: $6y = 1800$, $y = 300$ and $x = 600$. Use 600 kg of the 14% ration and 300 kg of the 26% ration ($84 + 78 = 162$ kg of protein, and $162 \div 900 = 18\%$ ✓). **13.** Let x = hours at 100 km/h and y = hours at 60 km/h: $x + y = 8.5$ and $100x + 60y = 750$. Substituting $y = 8.5 - x$: $40x + 510 = 750$, $x = 6$ and $y = 2.5$. She drove 6 h at 100 km/h and 2.5 h at 60 km/h ($600 + 150 = 750$ km ✓, $6 + 2.5 = 8.5$ h ✓). **14.** (a) Going downstream the current pushes the boat along, so the speeds add; going upstream the current pushes back, so the current's speed is subtracted. (b) $2(b + c) = 60$ gives $b + c = 30$; $3(b - c) = 60$ gives $b - c = 20$. Adding, $2b = 50$, $b = 25$ and $c = 5$. Still-water speed 25 km/h, current 5 km/h (downstream 30 km/h for 2 h $= 60$ km ✓; upstream 20 km/h for 3 h $= 60$ km ✓). **15.** $2l + 2w = 174$, so $l + w = 87$; with $l = 2w - 3$, $3w - 3 = 87$, $w = 30$ and $l = 57$. The rink is 57 m by 30 m, with area $57 \times 30 = 1710$ m^2 (perimeter $2(57 + 30) = 174$ m ✓, and $2(30) - 3 = 57$ ✓). **16.** (a) $C = 18n$ and $C = 120 + 6n$; setting them equal, $18n = 120 + 6n$, $12n = 120$, $n = 10$ and $C = 180$. At 10 visits both plans cost \$180. (b) Drop-in: $18(15) = \$270$; membership: $120 + 6(15) = \$210$. The membership is cheaper by \$60. (c) It is the point of intersection of the two lines, $(10, 180)$. **17.** Let t and b be seedlings per hour for Tia and Ben: $5t + 3b = 340$ and $2t + 6b = 280$. Halving the second gives $t + 3b = 140$; subtracting from the first, $4t = 200$, $t = 50$ and $b = 30$. Tia plants 50 seedlings per hour and Ben plants 30 (Monday $250 + 90 = 340$ ✓; Tuesday $100 + 180 = 280$ ✓). **18.** (a) Substituting $s = \ell + 60$ into $s - \ell = 60$ gives $(\ell + 60) - \ell = 60$, i.e. $60 = 60$, which is true for every ℓ — no single solution. His two equations are the same fact written two ways, so his system carries only one piece of information. (b) He never used the \$1080 in sales. A correct system is $s = \ell + 60$ and $5s + 8\ell = 1080$: $5(\ell + 60) + 8\ell = 1080$, $13\ell = 780$, $\ell = 60$ and $s = 120$. They sold 120 small and 60 large ($\$600 + \$480 = \$1080$ ✓, and 120 is 60 more than 60 ✓). **19.** (a) From $a + y = 120$, $y = 120 - a$; then $10(120 - a) - 15a = 1300$ gives $1200 - 25a = 1300$, $-25a = 100$, $a = -4$ and $y = 124$. A count of -4 adult tickets is impossible — you cannot sell a negative number of tickets. (b) She reversed the subtraction: the *adult* revenue $15a$ is the larger amount, so the correct equation is $15a - 10y = 1300$. Then $15a - 10(120 - a) = 1300$, $25a = 2500$, $a = 100$ and $y = 20$. They sold 100 adult and 20 youth tickets ($\$1500 - \$200 = \$1300$ ✓, $100 + 20 = 120$ ✓). **20.** (a) Sentence "26 boats were rented": $16 + 10 = 26$ ✓. Sentence "for a total of \$990": $45(16) + 30(10) = 720 + 300 = \$1020$, not \$990 (×). Jonah's answer satisfies the count but not the money, so it is wrong. (b) $c + k = 26$ and $45c + 30k = 990$; substituting $k = 26 - c$ gives $15c + 780 = 990$, $c = 14$ and $k = 12$. They rented 14 canoes and 12 kayaks. In words: $14 + 12 = 26$ boats ✓, and $45(14) + 30(12) = 630 + 360 = \$990$ ✓. **21.** (a) $6(85) + 11(55) = 510 + 605 = \$1115$. (The counts also check: $85 + 55 = 140$ passes.) (b) The dark roast costs $15(22) = \$330$, so the other $40 - 15 = 25$ kg must cost $680 - 330 = \$350$, which is $350 \div 25 = \$14$/kg. **22.** (a) The 620 containers from Day 1 is not needed — both facts you need (the two-day total count and the two-day total value) already cover the whole drive, so the one-day split is extra information. (b) $c + b = 1500$ and $0.10c + 0.25b = 255$; multiplying the second by 100 gives $10c + 25b = 25500$, i.e. $2c + 5b = 5100$. Substituting $c = 1500 - b$: $3000 + 3b = 5100$, $b = 700$ and $c = 800$. They collected 800 cans and 700 bottles ($800 + 700 = 1500$ containers ✓, and $\$80.00 + \$175.00 = \$255.00$ ✓). **23.** (a) $a + y = 240$ and $18a + 9y = 3384$; substituting $y = 240 - a$ gives $9a + 2160 = 3384$, $9a = 1224$, $a = 136$ and $y = 104$. Friday: 136 adult and 104 youth tickets ($\$2448 + \$936 = \$3384$ ✓). (b) Saturday sold $2(136) = 272$ adult and 104 youth tickets, for $18(272) + 9(104) = 4896 + 936 = \$5832$. (c) Over the two days $240 + 272 + 104 = 616$ tickets were sold.

Lesson 8 — Extra Practice

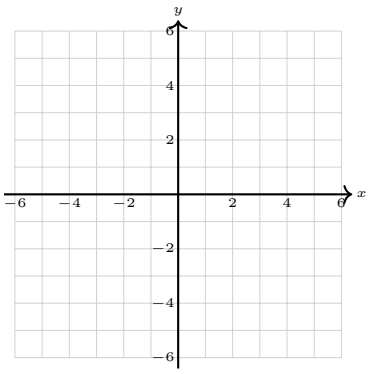
Unit Review: Systems of Linear Equations

This set is cumulative: every question pulls together two or more lessons. Keep the two summary tables from Lesson 8 beside you — the three methods and when each one suits, and the three outcomes (one solution, no solution, infinitely many) with the algebra that signals each.

Part A — One System, Three Methods

1. Consider the system $x + 2y = 8$ $y = x + 1$. Solve it **three separate ways**. Do not use an earlier answer to shortcut a later method — each route must stand on its own.

(a) **Graphing (L1).** Write both equations in slope-intercept form, graph them, and read the point of intersection off the grid.



(b) **Substitution (L3).** The second equation is already solved for y. Substitute into the *first* equation, solve for x, then back-substitute.

(c) **Elimination (L5).** Rewrite $y = x + 1$ in the form $-x + y = 1$, then add the two standard-form equations to eliminate x.

(d) State the solution as an ordered pair and check it in **both** original equations. Then answer: which of the three methods was fastest *for this system*, and what feature of the equations made it fastest? Name one feature that would have made a different method the best choice instead.

Part B — Choosing a Method Before You Solve

2. For each system, state which method you would use — graphing, substitution, or elimination — and justify it in one sentence by pointing at a specific feature of the equations (an isolated variable, a coefficient of 1 or -1, matching or opposite coefficients, or clean slope-intercept form with integer intercepts). **Do not solve.**

(a) $y = 3x - 4$
$2x + 5y = 7$

(b) $4x - 7y = 10$
$4x + 3y = -6$

(c) $y = -x + 5$
$y = 2x - 1$

(d) $3x + 2y = 12$
$5x - 4y = 2$

(e) $x - 6y = 11$
$8x + 3y = -4$

(f) $2x + 9y = 5$
$7x - 4y = 3$

3. Three students look at the system $6x - 5y = 4$ $2x + y = 9$.

(a) Ines says elimination here forces you to multiply *both* equations. Is she right? Show a single multiplication that eliminates a variable, or explain why none exists.

(b) Tavish says substitution is fastest. Point to the feature of the system that supports him, and say which variable he should isolate and from which equation.

(c) Ren graphs the two lines and reads the solution as $(3, 3)$. Solve the system exactly, then explain what went wrong with Ren's approach.

Part C — Solving by Every Method

4. Solve each system by **substitution**. In (c) and (d) you must isolate a variable first. Answers may be negative. Check each solution in both original equations.

(a) $y = 2x + 7, \quad 3x - 2y = -12$

(b) $x = 4 - 3y, \quad 2x + 5y = 3$

(c) $x - 2y = -9, \quad 4x + 3y = -14$

(d) $3x + y = 2, \quad 5x - 4y = 26$

5. Solve each system by **elimination**. In (a) and (d) one variable is already set up; in (b) and (c) you must multiply first. Two of these have a fractional coordinate — leave it as an exact fraction, never a rounded decimal.

(a) $2x + 5y = 1, \quad 2x - 3y = 17$

(b) $4x + 3y = -1, \quad 5x - 2y = 16$

(c) $3x - 7y = 16, \quad 5x + 2y = 13$

(d) $6x + 4y = 7, \quad 2x - 4y = 1$

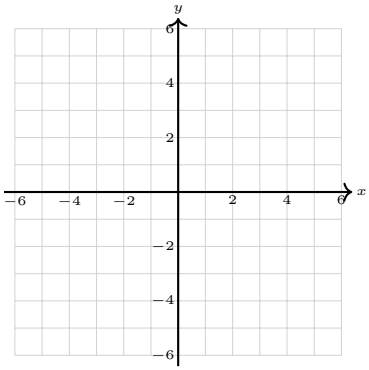
6. Now **you** choose. Solve each system by whichever method is most efficient, and write the name of the method you used beside your answer. One of the four is best done by graphing — decide which, and use the grid you are given for it.

(a) $y = -3x + 11, \quad 4x + 3y = 23$

(b) $7x + 2y = 23, \quad 7x - 5y = 16$

(c) $y = \frac{1}{2}x + 1, \quad y = -x + 4$

(d) $5x - 3y = -7, \quad 2x + 7y = 30$



Part D — Classifying and Reasoning

7. Without solving, state whether each system has one solution, no solution, or infinitely many. Some equations are in standard form, so rewrite them as $y = mx + b$ first and compare m, then b. Justify each answer with the slopes and intercepts.

(a) $y = -4x + 3$
$8x + 2y = 6$

(b) $3x - y = 5$
$6x - 2y = 7$

(c) $x + y = 4$
$x - y = 4$

(d) $2x + 6y = 9$
$y = -\frac{1}{3}x + 2$

(e) $5x - 2y = 8$
$y = \frac{5}{2}x - 4$

(f) $4x + y = 9$
$x - y = 1$

8. Solve each system algebraically using substitution or elimination. **Not every system has a single point as its answer.** If the variables vanish, do not write "no answer": state whether the statement you are left with is true or false, say what that means about the solution, and describe the graph in words.

(a) $y = 2x - 5, \quad 6x - 3y = 15$

(b) $x - 4y = 6, \quad 3x - 12y = 10$

(c) $2x + 3y = 12, \quad 4x + 5y = 22$

(d) $3x - 6y = 9, \quad -x + 2y = -3$

9. **Finding the missing constant.** In each part, determine the value of k that produces the stated outcome, then *verify* by putting your k into the system and running substitution or elimination to the end — your work should finish with the matching signal ($x =$ a number, a false statement, or a true statement).

(a) For what value of k does $y = kx - 4,\ y = 6x + 1$ have **no solution**? Explain why no value of k gives infinitely many.

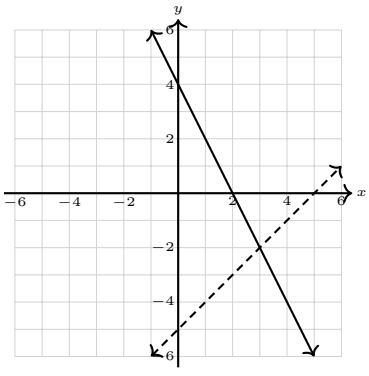
(b) For what value of k does $2x + ky = 10,\ x + 3y = 5$ have **infinitely many** solutions? Verify by elimination.

(c) For what values of k does $kx - 4y = 12,\ 3x + 2y = 5$ have **exactly one** solution? State the one excluded value and say what happens there.

(d) For what value of k does $3x + 5y = k,\ 6x + 10y = 24$ have infinitely many solutions? What is the outcome for *every other* value of k?

Part E — Graphs and Systems

10. The system $2x + y = 4$ (solid) and $x - y = 5$ (dashed) is graphed below. The lines cross at integer coordinates.

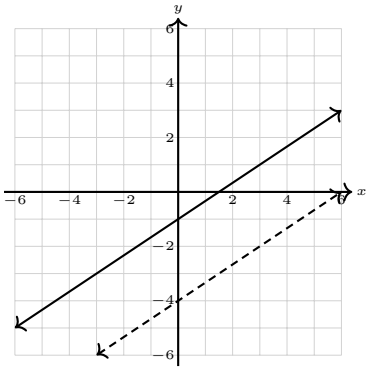


(a) Read the solution from the graph and write it as an ordered pair.

(b) Confirm it by **elimination**: both equations are in standard form and the y coefficients are opposites, so one step does it. Show the addition.

(c) Explain why part (b) is the proof and part (a) is only evidence.

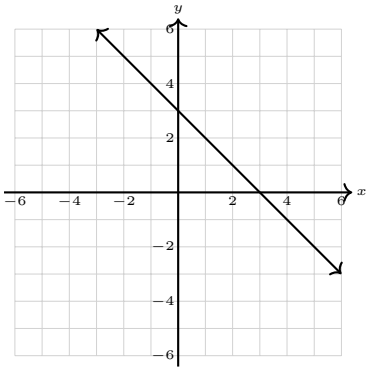
11. The system $y = \frac{2}{3}x - 1$ (solid) and $2x - 3y = 12$ (dashed) is graphed below.



(a) Classify the system from the picture, and support your answer using the slopes and intercepts.

(b) Confirm algebraically: multiply the first equation by 3, write it in standard form, and eliminate. State the statement you finish with and why it settles the question.

12. The line $y = -x + 3$ is drawn on the grid below. Write the equation of a *second* line so that the system made from the two lines has each outcome listed.



(a) No solution.
Equation: _____
Justify using m and b.

(b) Infinitely many solutions, written in *standard form* so it does not simply repeat $y = -x + 3$.
Equation: _____

(c) Exactly one solution, at the point $(2, 1)$.
Equation: _____
Graph your line on the grid, then check $(2, 1)$ in both equations.

Part F — Error Analysis

13. Marcus solves $2x - 6y = 4$ and $-x + 3y = -2$ by multiplying the second equation by 2 and adding. He reaches $0 = 0$ and writes "no solution, because there is no x and no y."

(a) Reproduce his elimination and confirm that $0 = 0$ really is what the algebra gives.

(b) Explain why $0 = 0$ is a *true* statement and what that means. State the correct classification, support it by writing both equations as $y = mx + b$, and describe the graph.

14. Kofi solved $4x + 3y = 11$ and $4x - 5y = 3$ by elimination. **Exactly one** of his steps is wrong.
Step 1: Both equations start with $4x$, so subtract the second from the first.
Step 2: $3y - 5y = 11 - 3$, so $-2y = 8$.
Step 3: $y = -4$.
Step 4: $4x + 3(-4) = 11$, so $4x = 23$ and $x = \frac{23}{4}$.
Step 5: The solution is $\left(\frac{23}{4}, -4\right)$.

(a) Substitute Kofi's answer into the *second* original equation to show it fails.

(b) Identify the numbered step that contains the error and describe the mistake precisely.

(c) Redo the work correctly and state the true solution.

15. Priya solved $y = 3x - 4$ and $2x + 5y = 14$ by substitution. Again exactly one step is wrong.
Step 1: $2x + 5(3x - 4) = 14$.
Step 2: $2x + 15x - 4 = 14$.
Step 3: $17x = 18$, so $x = \frac{18}{17}$.
Step 4: $y = 3\left(\frac{18}{17}\right) - 4 = -\frac{14}{17}$.

(a) Name the step with the error and state the rule she broke.

(b) Solve correctly, then check your point in both original equations.

Part G — Applications

16. Two shops will print team shirts. Shop A charges a \$60 setup fee plus \$4 per shirt; Shop B charges a \$20 setup fee plus \$6 per shirt. Let n be the number of shirts and C the total cost.
(a) Write a system of two equations, one for each shop.
(b) Both equations are solved for C, so name the method this fact makes obvious, and use it to find the break-even number of shirts and the cost there.
(c) Interpret the solution in a sentence about the two shops.
(d) A team needs 30 shirts. Which shop is cheaper, and by how much? Answer (d) using your equations, not a new guess.

17. A lab technician mixes a 20% acid solution with a 50% acid solution to make 30 L of a 40% acid solution. Let x be the litres of 20% solution and y the litres of 50% solution.
(a) Write one equation for the total *volume* and one for the total *acid*. Multiply the acid equation by 10 to clear the decimals.
(b) Solve, naming the method and why the volume equation makes it the natural choice.
(c) Check your answer by computing the litres of pure acid in each container and in the mixture.

18. A canteen cash box holds 52 coins, all of them quarters (\$0.25) and loonies (\$1.00), worth \$34.75 in total.
(a) Define your two variables in words, then write a *count* equation and a *value* equation. Multiply the value equation by 4 to clear the decimals.
(b) Solve, and answer the question in a full sentence.
(c) Check both the coin count and the dollar total.

19. A bakery sold 140 items in a morning: muffins at \$2.75 each and scones at \$3.50 each, for total sales of \$432.25.
(a) Set up a system. (b) Solve it by whichever method you prefer, and justify the choice in one sentence. (c) Verify both the item count and the money.

Part H — Challenge

20. **Work backwards.** The system $ax + 3y = 12$ and $5x - by = 1$ is known to have the solution $(3, 2)$.
(a) Use the fact that $(3, 2)$ satisfies each equation to find a and b. Note that each unknown appears in only one equation, so you never need a system to find them.
(b) Write out the completed system and solve it from scratch by elimination. Your answer must come back to $(3, 2)$ — if it does not, recheck part (a).

21. **Literal coefficients.** Solve the system $2x + y = 7a$ and $x - y = 2a$ for x and y in terms of the constant a.
(a) Add the equations to eliminate y, then back-substitute. Your answers will contain a.
(b) Substitute $a = 4$ into your general answer, then solve the numerical system $2x+y = 28$, $x-y = 8$ independently. Confirm the two agree.

22. **Three lines.** Consider $\ell_1 :\ x + y = 5$, $\ell_2 :\ 2x - y = 1$, $\ell_3 :\ 3x + 2y = 12$.
(a) Solve ℓ_1 with ℓ_2, then test that point in ℓ_3. Do all three lines pass through one common point?
(b) Now replace ℓ_3 with $\ell_3' :\ 3x + 2y = 13$. Show that each *pair* of lines still meets, but that no point lies on all three. Describe the picture the three lines make.

23. **Build one.** Write a system of two equations in standard form whose solution is $(-3, 4)$ and which **cannot** be solved by multiplying only one equation — eliminating either variable must require multiplying *both*. Explain how you guaranteed that, then solve your own system by elimination to prove it gives $(-3, 4)$.

24. **Clear first, then solve.** Neither system below is ready for elimination. Clear the fractions or decimals from each equation first, then solve, then check your point in the *original* fractional or decimal equations.

(a) $\dfrac{x}{2}+\dfrac{y}{3}=4, \qquad \dfrac{x}{4}-y=-5$ (b) $0.4x+0.3y=2.7, \qquad 0.2x-0.5y=-1.9$

Answer Key. 1. (a) $y=-\frac{1}{2}x+4$ and $y=x+1$ cross at $(2,3)$. (b) $x+2(x+1)=8 \Rightarrow 3x=6 \Rightarrow x=2$, $y=3$. (c) $x+2y=8$ plus $-x+y=1$ gives $3y=9$, $y=3$, $x=2$. (d) $(2,3)$; check $2+2(3)=8$ ✓ and $3=2+1$ ✓. Substitution was fastest because y was already isolated, so one substitution finished it. Graphing would have been best if both equations had been given as $y=mx+b$ with integer intercepts; elimination would have been best if both had been in $Ax+By=C$ with matching or opposite coefficients. **2.** (a) Substitution — y is already isolated. (b) Elimination — the x coefficients match at 4, so subtracting kills x in one step. (c) Graphing, or setting the two expressions equal — both are in $y=mx+b$ with integer intercepts and they meet at the lattice point $(2,3)$. (d) Elimination — multiply the first by 2 to get $\pm 4y$; only one equation needs multiplying. (e) Substitution — x has coefficient 1 in the first equation, so $x=6y+11$ costs nothing. (f) Elimination after multiplying *both* — no coefficient is 1 and nothing matches (use ×7 and ×2, or ×4 and ×9). **3.** (a) No, she is wrong: multiplying only the second equation by 5 gives $10x+5y=45$, which added to $6x-5y=4$ eliminates y. (b) y has coefficient 1 in $2x+y=9$, so isolate y there: $y=9-2x$. (c) $6x-5(9-2x)=4 \Rightarrow 16x=49$, so $x=\frac{49}{16}$, $y=\frac{23}{8}$. The intersection is not at integer coordinates, so Ren's graph could only give a nearby-looking guess; a graph is unreliable unless the crossing is a lattice point. **4.** (a) $(-2,3)$ (b) $(-11,5)$ (c) $(-5,2)$ (d) $(2,-4)$. **5.** (a) subtract: $8y=-16$, $y=-2$, $x=\frac{11}{2}$, so $\left(\frac{11}{2},-2\right)$ (b) $(2,-3)$ (c) $(3,-1)$ (d) add: $8x=8$, $x=1$, $y=\frac{1}{4}$, so $\left(1,\frac{1}{4}\right)$. **6.** (a) $(2,5)$, substitution (y isolated) (b) $(3,1)$, elimination (matching $7x$, subtract) (c) $(2,2)$, graphing (both $y=mx+b$, lattice crossing) (d) $(1,4)$, elimination after multiplying both (×7 and ×3). **7.** (a) Infinitely many — $8x+2y=6$ is $y=-4x+3$, the same line. (b) No solution — both slopes 3, intercepts -5 and $-\frac{7}{2}$. (c) One solution — slopes -1 and 1; the point is $(4,0)$. (d) No solution — both slopes $-\frac{1}{3}$, intercepts $\frac{3}{2}$ and 2. (e) Infinitely many — both are $y=\frac{5}{2}x-4$. (f) One solution — slopes -4 and 1; the point is $(2,1)$. **8.** (a) $6x-3(2x-5)=15$ gives $15=15$, true ⇒ infinitely many solutions; one line drawn twice. (b) $x=4y+6$ gives $18=10$, false ⇒ no solution; two parallel lines. (c) One solution, $(3,2)$. (d) Multiplying the second by 3 and adding gives $0=0$, true ⇒ infinitely many solutions; the two equations describe the same line $y=\frac{1}{2}x-\frac{3}{2}$. **9.** (a) $k=6$ (equal slopes, unequal intercepts -4 and 1); no k gives infinitely many because the intercepts -4 and 1 can never be made equal — k only controls the slope. Check $k=6$: $6x-4=6x+1 \Rightarrow -4=1$, false. (b) $k=6$; then $2x+6y=10$ is exactly twice $x+3y=5$, and subtracting $2\times(x+3y=5)$ leaves $0=0$. (c) One solution for every k except $k=-6$. At $k=-6$ the system is $-6x-4y=12$ with $3x+2y=5$; adding $2\times(3x+2y=5)$ gives $0=22$, false — no solution (parallel, both slope $-\frac{3}{2}$). (d) $k=12$ gives infinitely many (the second equation is twice the first); every other k gives no solution, since the slopes always match at $-\frac{3}{5}$ but the intercepts differ. **10.** (a) $(3,-2)$ (b) adding gives $3x=9$, $x=3$, then $3-y=5$ so $y=-2$ (c) The graph only shows where the lines *appear* to meet and depends on the accuracy of the drawing; the elimination is exact arithmetic, and checking $(3,-2)$ in both originals proves it. **11.** (a) No solution — the lines are parallel; both have slope $\frac{2}{3}$ but intercepts -1 and -4. (b) $3y=2x-3$, so $2x-3y=3$; subtracting this from $2x-3y=12$ gives $0=9$, a false statement, which is the algebraic signal for no solution. **12.** Sample answers. (a) Any line with slope -1 and a different intercept, e.g. $y=-x-2$: same m, different b ⇒ parallel. (b) $2x+2y=6$ (or $3x+3y=9$) — dividing by 2 returns $y=-x+3$, the same line. (c) Any line through $(2,1)$ whose slope is not -1, e.g. $y=x-1$: check $1=-2+3$ ✓ and $1=2-1$ ✓. Note $(2,1)$ must lie on the given line, and it does. **13.** (a) $2\times(-x+3y=-2)$ is $-2x+6y=-4$; adding it to $2x-6y=4$ gives $0=0$. (b) $0=0$ is always true, so *every* point on the line satisfies both equations: infinitely many solutions, not none. Both equations become $y=\frac{1}{3}x-\frac{2}{3}$, so the graph is one line drawn twice. "No solution" is the $0=8$ case, a *false* statement. **14.** (a) $4\left(\frac{23}{4}\right)-5(-4)=23+20=43\neq 3$. (b) Step 2: subtracting the second equation must change the sign of *every* one of its terms, so the y terms give $3y-(-5y)=8y$, not $-2y$. (c) $8y=8$, $y=1$, then $4x+3=11$ so $x=2$; the solution is $(2,1)$. **15.** (a) Step 2 — she distributed the 5 over $3x$ but not over -4; $5(3x-4)=15x-20$. (b) $2x+15x-20=14 \Rightarrow 17x=34 \Rightarrow x=2$, $y=3(2)-4=2$, so $(2,2)$; check $2=3(2)-4$ ✓ and $2(2)+5(2)=14$ ✓. **16.** (a) $C=4n+60$, $C=6n+20$. (b) Both are solved for C, so substitute one into the other (equate them): $4n+60=6n+20 \Rightarrow 40=2n \Rightarrow n=20$ and $C=\$140$. (c) At 20 shirts both shops charge \$140; below 20 shirts Shop B is cheaper, above 20 Shop A is cheaper. (d) $n=30$: Shop A = \$180, Shop B = \$200, so Shop A is cheaper by \$20. **17.** (a) $x+y=30$ and $0.2x+0.5y=12$, i.e. $2x+5y=120$. (b) Substitution or elimination; $x+y=30$ has coefficients of 1, so $x=30-y$ (or double it and subtract) is quickest: $x=10$ L of 20% and $y=20$ L of 50%. (c) Acid: $0.2(10)+0.5(20)=2+10=12$ L, and $0.4(30)=12$ L ✓; volume $10+20=30$ L ✓. **18.** (a) $q=$ quarters, $\ell=$ loonies; $q+\ell=52$ and $0.25q+\ell=34.75$, i.e. $q+4\ell=139$. (b) Subtracting gives $3\ell=87$, so $\ell=29$ and $q=23$: the box holds 23 quarters and 29 loonies. (c) $23+29=52$ coins ✓; $23(0.25)+29(1.00)=5.75+29.00=\$34.75$ ✓. **19.** (a) $m+s=140$, $2.75m+3.50s=432.25$ (multiply by 4: $11m+14s=1729$). (b) $m=77$ muffins and $s=63$ scones — elimination is tidy once the money equation is cleared, and substitution works too since $m+s=140$ has coefficients of 1. (c) $77+63=140$ ✓; $77(2.75)+63(3.50)=211.75+220.50=\$432.25$ ✓. **20.** (a) $3a+6=12 \Rightarrow a=2$; $15-2b=1 \Rightarrow b=7$. (b) $2x+3y=12$ and $5x-7y=1$; ×7 and ×3 give $14x+21y=84$ and $15x-21y=3$, so $29x=87$, $x=3$, $y=2$ — back to $(3,2)$ ✓. **21.** (a) Adding: $3x=9a$, so $x=3a$; then $3a-y=2a$ gives $y=a$. Check: $2(3a)+a=7a$ ✓, $3a-a=2a$ ✓. (b) $a=4$ gives $(12,4)$; solving $2x+y=28$, $x-y=8$ directly also gives $(12,4)$ — they agree. **22.** (a) ℓ_1 and ℓ_2 meet at $(2,3)$, and $3(2)+2(3)=12$ ✓, so yes, all three lines pass through $(2,3)$ (the three-equation system is consistent). (b) With ℓ_3' : $3x+2y=13$: $\ell_1\cap\ell_2=(2,3)$, $\ell_1\cap\ell_3'=(3,2)$, and $\ell_2\cap\ell_3'=\left(\frac{15}{7},\frac{23}{7}\right)$ — three different points, so no point satisfies all three and the three-equation system has no solution. The lines form a small triangle. **23.** Sample: $2x+3y=6$ and $5x-4y=-31$. Check: $2(-3)+3(4)=6$ ✓, $5(-3)-4(4)=-31$ ✓. It is guaranteed because no coefficient is 1 or -1 and neither column divides evenly — for x you need ×5 and ×2, for y you need ×4 and ×3. Solving: $10x+15y=30$ and $10x-8y=-62$, subtract ⇒ $23y=92$, $y=4$, then $2x+12=6$ gives $x=-3$. (Any system meeting the two conditions is acceptable.) **24.** (a) ×6 and ×4: $3x+2y=24$ and $x-4y=-20$; solving gives $(4,6)$. Check: $\frac{4}{2}+\frac{6}{3}=2+2=4$ ✓ and $\frac{4}{4}-6=-5$ ✓. (b) ×10: $4x+3y=27$ and $2x-5y=-19$; solving gives $(3,5)$. Check: $0.4(3)+0.3(5)=1.2+1.5=2.7$ ✓ and $0.2(3)-0.5(5)=0.6-2.5=-1.9$ ✓.